Guide Before Commitment: When Does a Diffusion Model Decide What It Draws, and What Does Classifier-Free Guidance Change?

Author names and affiliations: *placeholder*

Pre-registered study; development split complete, test split partial (6 of 9 shards)

Abstract

Text-to-image diffusion models often violate part of a compositional prompt. We ask when, along the denoising trajectory, each kind of constraint (object presence, count, colour binding, spatial relation, relative size) stops depending on the remaining randomness, and what classifier-free guidance (CFG) changes about that timing. We measure it interventionally: saved trajectory states of Stable Diffusion 1.5 are re-noised and re-denoised, and the agreement of outcomes, corrected for chance agreement between independent seeds, gives a persistence curve $C(t)$ and its half-point t_{50}. Hypotheses, decision rules and development-set constants were frozen in a signed protocol before any test image existed. A human audit validated the automatic judge only for object presence. Six of nine test shards were generated, so all test results are partial and exploratory. The one registered rule that passed concerns guidance: stronger CFG makes object presence persistent earlier ($t_{50}(w{=}15) - t_{50}(w{=}2) = -8.67$ of 40 steps, 95% CI $[-16.34, -4.72]$, Holm $p = 0.046$). The ordering is not an artefact of the chance correction, but an independent-judge-error stress test at the audited error rate removes it. Cross-attention shows the same direction: on identical initial noise, higher CFG separates object nouns and their attention stabilises earlier, and objects that are later missed already show diffuse, overlapping attention in the first steps. A guidance schedule timed by the development-set measurement (Commitment-Aligned Guidance) failed its registered test. Its switch steps did not transfer to the test scenes, and matched-mean controls attribute its diversity gain to lower average guidance rather than to timing. Commitment times also did not replicate across splits, and a full-resample control revealed a presence offset that we characterise but cannot yet explain.

1 Introduction

Prompt adherence of text-to-image diffusion models is usually summarised by one score per image. Compositional failures, however, are constraint-specific: an object can be present while its colour is wrong, or two objects can be correct while their left–right order is not. If different constraints are resolved at different moments of the sampling trajectory, three practical consequences follow. Guidance strength that matters early may be wasted late; diagnostics that read early internal signals are forecasts for some constraints and read-outs for others; and a pooled adherence score can hide both effects.

Prior work establishes that denoising stages control different image properties [1, 17, 23, 31], that guidance can be restricted or scheduled over time [16, 22, 27], and that cross-attention carries information about objects and attributes that can be read and steered [3, 5, 10, 29]. We therefore do not claim that the general idea of stage-dependent generation is new. Our questions are narrower and measurement-first:

1. **When** does each constraint type (presence, count, colour binding, spatial relation, relative size) stop depending on the remaining randomness of the trajectory? We answer with an interventional, chance-corrected persistence curve rather than with an observational signal.

2. **What does CFG change?** Does a larger guidance scale move persistence earlier, once the higher seed-to-seed agreement it induces is accounted for?

3. **Can attention tell?** Does the step at which cross-attention stops changing track the interventional measurement?

4. **Is the timing usable?** Does a two-level guidance schedule whose switch step is read off development-set measurements keep adherence while raising diversity, and does it beat the same mean guidance applied in reverse order?

Contributions.

(i) An interventional, chance-corrected measurement of restart persistence per constraint type, with two re-noising probes and a closed-form stress test for judge error (Section 4).

(ii) A controlled scene-graph benchmark with counterfactual variants and a factorial size family, scored by a judge whose blinded human audit decides which constraint types may carry claims (Sections 5–6).

(iii) Commitment-Aligned Guidance (CAG), tested against constant, reversed and matched-mean schedules under a signed pre-registration, with every gate reported whether it passed or failed (Sections 4 and 8).

(iv) On the available test shards, an account of what survives: the guidance effect on timing, its sensitivity to the judge, the failure of commitment times to transfer, the dose explanation of CAG's diversity gain, and what saved cross-attention maps add to the restart measurement (Sections 9–11).

Findings in brief (partial test split, exploratory).

- Stronger guidance makes object presence persistent earlier, with $t_{50}(w{=}15) - t_{50}(w{=}2) = -8.67$ of 40 steps. This is the only registered rule that passed. The effect is not an artefact of the chance correction or of the full-resample offset. It does assume that the presence judge's errors are not independent across seeds.

- Cross-attention moves in the same direction: higher guidance separates the object nouns and makes their attention stabilise earlier. Objects that are eventually missed have diffuse, overlapping attention from the first steps.

- Attention stabilisation does not order the constraint types the way persistence does. Its binding signal follows object layout, not colour assignment.

- Commitment times measured on the development split did not transfer to the test scenes, so CAG switched guidance down before binding had settled. CAG was not non-inferior to constant guidance, and its diversity gain is explained by its lower mean guidance.

We use the operational term *restart persistence*; we do not claim that the model "decides" in any cognitive sense. Figure 1 summarises the probes and the schedule.

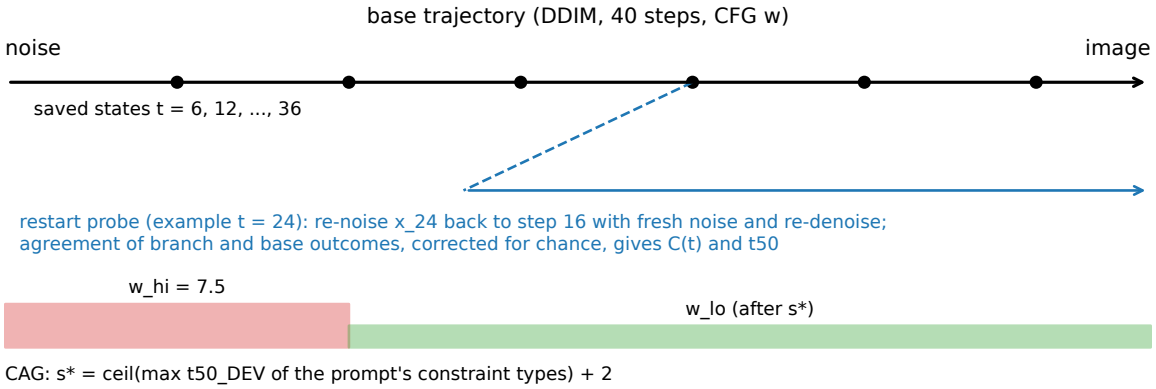


Figure 1: **Conceptual diagram** (not data). Top: a 40-step DDIM trajectory with saved states. Middle: the restart probe re-noises a saved state to an earlier step with fresh noise and re-denoises; agreement of branch and base outcomes, corrected for chance agreement between independent seeds, gives $C(t)$ and t_{50}. Bottom: CAG applies w_{hi} until a prompt-specific switch step $s^\star$ derived from development-set t_{50}, then w_{lo}.

2 Related work

Diffusion sampling and guidance. We use denoising diffusion models [12], deterministic DDIM sampling [28], latent diffusion [26] and classifier-free guidance [11]. Restricting guidance to an interval of noise levels has been reported to improve sample quality [16], annealing the conditioning signal early to restore diversity [27], and choosing per-step guidance online with latent evaluators to be prompt-dependent [22]. Add-noise-then-denoise operations are used for editing and for sampling [19, 30]; we use re-noising only as a *measurement*.

Stages of generation. Switching prompts over denoising intervals localises layout, shape and detail to successive stages [23]; stage-specific text embeddings control different attributes [31]; an ensemble of stage-specialised denoisers is motivated by text conditioning mattering most early [1]; and features have been analysed theoretically as emerging in narrow time windows [17]. Strengthening the conditioning signal at selected steps has been reported to improve alignment more when done late than early [8], a finding in tension with front-loading guidance, and negative prompts to act with a delay [2].

Cross-attention. Cross-attention maps are used to interpret [29] and edit [10] generations, and attention-based objectives target object neglect and attribute binding [3, 6, 18, 25]; attention can also read and steer object size and position [5]. Initial-noise search and early prediction of image quality from attention are close in aim to our attention analysis [4, 9].

Evaluation. Object-focused and question-answering evaluators [7, 13–15] motivate our emphasis on judge reliability. Our judge combines open-vocabulary detection [20] with CLIP [24]; seed diversity uses DINOv2 embeddings [21].

Position of this work. Table 1 contrasts what the closest methods measure, intervene on and evaluate. The common thread is that timing matters; the differences are that we (a) measure timing

Table 1: Closest prior work and this study: what each method measures, what it changes during sampling, and how it is evaluated. "–": not an aim of the method. Descriptions of Prompt-to-Prompt and Attend-and-Excite were checked against the full papers, the others against their abstracts (marked in the bibliography source).

Method	Measures	Intervenes on	Evaluated by
Prompt-to-Prompt [10]	–	cross-attention maps (injection, re-weighting) to edit an image	edit fidelity, qualitative and quantitative
Attend-and-Excite [3]	subject-token attention as a neglect signal	latents at inference, to raise neglected tokens' attention	object presence and binding (CLIP-based metrics, user study)
Prompt Mixing [23]	–	which prompt is used in which denoising interval	localisation of shape variations
Critical windows [17]	time windows in which features emerge (theory)	–	theory and experiments
Interval guidance [16]	–	CFG applied only in a limited noise interval	distribution and sample quality
CADS [27]	–	conditioning annealed with noise early in sampling	diversity and quality
Dynamic CFG [22]	per-step feedback from latent evaluators	per-step guidance scale, chosen online	prompt adherence and quality
Step selection [8]	–	conditioning signal strengthened at selected steps	text–image alignment
InitNO [9]	attention-based validity of the initial noise	initial noise, optimised before sampling	text–image alignment
Diffusion Probe [4]	early cross-attention as a predictor of final quality	– (used for prompt and seed selection)	prediction accuracy
This study	interventional, chance-corrected persistence $C(t)$ and t_{50} per constraint type; attention stabilisation	two-level CFG switched at a step derived from development-set t_{50} (CAG)	pre-registered tests with a human-audited judge; constant, reversed and matched-mean schedule controls

interventionally and per constraint type with an explicit chance correction, (b) test whether CFG moves that timing, (c) compare an unsupervised attention signal against the interventional one, and (d) derive a guidance switch from the measurement rather than tuning it on adherence. To our knowledge these four elements have not been combined on controlled scene graphs; the literature search was narrow (Table 1 caption) and this is a candidate gap, not an established one.

3 Hypotheses and pre-registration

The four hypotheses form one Holm family ($\alpha = 0.05$). A hypothesis is reported *supported* only if its own decision rule passes *and* its Holm-adjusted $p < 0.05$, with Holm computed over all four hypotheses even when one is not estimable. Definitions, decision rules and development-set constants were fixed in a protocol addendum and a signed protocol file before any test image was generated (103 code, configuration, benchmark and documentation files hashed; Section 15).

Table 2: Status of each hypothesis fixed before the test split (part of the signed protocol; applied automatically by the analysis code).

Hyp.	Registered rule	Status for the test split
HA	Δt_{50} rule + Holm	EXPLORATORY: DEV precision gate failed (half-width 7.94 > 4); binding judge below criterion
HD	$\Delta t_{50}^{\mathrm{CFG}}$ rule + Holm	confirmatory through presence only; binding estimand exploratory
HE	intersection–union + Holm	registered rule unchanged; F includes judge types below the criterion (disclosed); presence-only F as exploratory sensitivity
HF	concordance/Spearman rule + Holm	registered rule unchanged; type orderings include judge types below the criterion (disclosed)

HA, constraint-specific persistence. Primary estimand $\Delta t_{50} = t_{50}^{\mathrm{bind|pres}} - t_{50}^{\mathrm{pres}}$ at $w = 7.5$ with the fixed-jump restart (D8). Support: the 95% scene-bootstrap interval excludes 0, $|\Delta t_{50}| \geq 3$ steps, and the joint-binding companion and the equal-ratio restart (ER30) agree in sign. Bounded null: the interval lies inside $(-1.5, 1.5)$.

HD, CFG moves persistence. $\Delta t_{50}^{\mathrm{CFG}} = t_{50}(w{=}15) - t_{50}(w{=}2)$ for presence and for binding given presence, with chance agreement computed separately at each w and one shared scene resample. Supported if at least one estimand is negative with $|\Delta| \geq 3$ and an interval excluding 0; $p_{\mathrm{HD}} = \min(1, 2 \min p)$. A positive interval is reported as contradicting HD.

HF, attention stabilisation. Supported if the ordering of constraint types by t_{stab} is concordant with their ordering by t_{50} (concordance ≥ 0.8) and the within-type Spearman correlation between scene t_{stab} and the scene's area under $C(t)$ has a 95% interval above 0 for at least two types. Rejected if concordance is at most 0.5, or if every within-type interval contains 0 with upper bound below 0.15; otherwise inconclusive.

HE, Commitment-Aligned Guidance. Supported only if all three hold (intersection–union, p = maximum of the three one-sided p): adherence F of CAG (A2) is non-inferior to constant CFG 7.5 (A1) (90% interval lower bound > -0.03), seed diversity is higher (95% interval > 0), and A2 beats the reversed schedule A4 with equal mean guidance on F (95% interval > 0).

Status fixed before the test split. Development-set evidence (Section 8) fixed the following before any test image existed (Table 2): HA is exploratory because its precision gate failed; per-type results for count, binding, spatial relation and relative size are exploratory because their judge accuracy missed the audit criterion; HD can be supported only through its presence estimand; HE and HF keep their registered rules, and their reports disclose that F and the type orderings include judge types below the criterion.

4 Methods

Generator. Stable Diffusion 1.5 [26] at a pinned revision in fp16, DDIM [28] with $\eta = 0$ and $T = 40$ steps at 512^2 pixels, empty negative prompt. With text condition c and guidance scale w, the noise prediction at step i is the classifier-free combination [11]

$$\hat{\varepsilon}_i = \varepsilon_\theta(x_i, \varnothing) + w_i \left(\varepsilon_\theta(x_i, c) - \varepsilon_\theta(x_i, \varnothing)\right), \tag{1}$$

where w_i is constant ($w_i = w$) except under the guidance schedules below. When $w_i = 1$ only the conditional branch is evaluated, and the number of network evaluations is counted per image.

Outcomes. An automatic judge (Section 6) scores each image for each constraint κ of its prompt: presence (all required objects detected), count (exact), and, conditional on the relevant objects being present, colour binding, left/right relation and relative size (box area of the intended larger object at least 1.5× the other's). Conditional outcomes are undefined, not zero, when an object is missing. The adherence score F of an image is the mean of the outcome families present.

Restart probe. The latent after $t \in \mathcal{G} = \{6, 12, 18, 24, 30, 36\}$ completed steps is saved. A branch re-noises it to an earlier step index j with fresh noise $\varepsilon \sim \mathcal{N}(0, I)$ and denoises deterministically to the end at the base's guidance:

$$x'_j = \sqrt{r}\, x_t + \sqrt{1 - r}\, \varepsilon, \qquad r = \bar{\alpha}(j)/\bar{\alpha}(t), \tag{2}$$

with $\bar{\alpha}(s)$ taken at the scheduler timestep of step s. The fixed-jump probe (D8) uses $j = \max(t - 8, 0)$, whose noise share $1 - r$ grows with t; the equal-ratio probe (ER30) chooses j so that $r \approx 0.30$ at every t; a full resample (FULL) restarts from pure noise and serves as a validity control.

Chance-corrected persistence. For scene s and constraint type κ, let $A_{\kappa s}(t)$ be the share of (base, branch) pairs with equal outcomes and $A0_{\kappa s}$ the agreement between independent seeds of the *same base prompt, guidance and step count.* Persistence pools scenes as a ratio estimator,

$$C_\kappa(t) = \frac{\sum_s \left[A_{\kappa s}(t) - A0_{\kappa s}\right]}{\sum_s \left[1 - A0_{\kappa s}\right]}, \tag{3}$$

so $C = 0$ means no more agreement than an independent sample and $C = 1$ means the outcome is fixed. t^κ_{50} is the linearly interpolated first crossing of 0.5; it is left-censored when C already exceeds 0.5 at $t = 6$ and right-censored when it never does. Because the noise share of D8 varies with t, absolute t_{50} is relative to the probe; ER30 is the registered sensitivity check.

Prompt-switch probe. A counterfactual trajectory (colour swap, relation reversal or size swap) is continued from its saved state with the canonical prompt, and

$$C^{\text{switch}}(t) = \frac{\sum_s \left[W_s(t) - W0_s\right]}{\sum_s \left[W1_s - W0_s\right]}, \tag{4}$$

where W is the share of switched images satisfying the counterfactual constraint and $W0$, $W1$ are that share among canonical and counterfactual images.

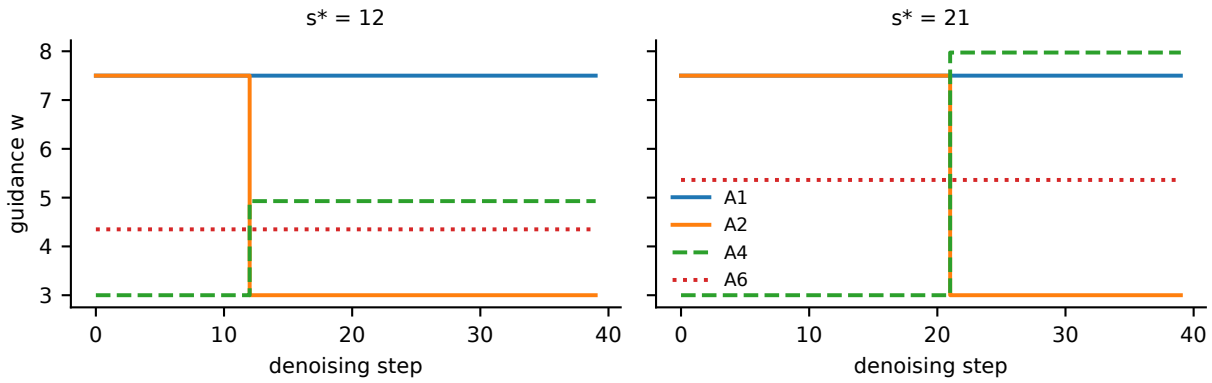


Figure 2: Exact guidance schedules of the frozen design for $s^\star = 12$ and $s^\star = 21$ with $w_{lo} = 3$: constant CFG (A1), CAG (A2), reversed with equal mean (A4) and constant at A2's mean (A6). A4 is not capped; its maximum guidance is recorded per job.

Attention stabilisation. Cross-attention maps of the constraint words are recorded at every step (conditional branch, averaged over heads and over layers within the 16×16 and 32×32 groups; recording leaves the image unchanged). Per type a signal a_i is formed (noun map, colour-word map, ratio of effective areas, angle between noun centroids, connected components), its velocity $v_i = 1 - \cos(a_i, a_{i-1})$ (or the absolute change of a scalar), and t_{stab} is the step at which half of the cumulative path $\sum_i v_i$ has been travelled.

Commitment-Aligned Guidance. For a prompt with constraint types K, the switch step and the schedules are

$$s^\star = \min\Big(\big\lceil \max_{\kappa \in K} t^{\kappa}_{50,\text{DEV}} \big\rceil + 2,\ T\Big), \tag{5}$$

$$w_i^{\text{A2}} = \begin{cases} w_{hi} & i < s^\star \\ w_{lo} & i \geq s^\star \end{cases}, \qquad w_i^{\text{A4}} = \begin{cases} w_{lo} & i < s^\star \\ w_{lo} + \frac{s^\star (w_{hi} - w_{lo})}{T - s^\star} & i \geq s^\star \end{cases}, \tag{6}$$

with $w_{hi} = 7.5$. The maximum runs over the prompt's constraint types (presence, count, binding, spatial relation, size), not over derived outcomes such as joint binding. A left-censored DEV value enters as the first grid step. A type without a DEV estimate would set $s^\star = T$, making CAG equal to constant guidance; this does not occur, because the size family carries no CAG arms. With the frozen DEV values (presence 9.3, binding 6.0 left-censored, spatial 6.0, count 18.9), every presence, binding and spatial prompt has $s^\star = 12$ and every prompt with a count has $s^\star = 21$; this was verified on every generated test CAG job. A4 has the same mean guidance as A2 in reverse order; A6 applies that mean as a constant; A3 uses a global $s^\star$ from the median DEV t_{50}; A1 is constant 7.5. The frozen design uses $w_{lo} = 3$, selected by the DEV quality guard (Section 8); Figure 2 shows the exact schedules for the two switch steps that occur on the test split.

Statistical analysis. Uncertainty uses a scene-level bootstrap with 10,000 draws and fixed random keys, resampling scenes (not images) so that seeds and branches of a scene stay together; p-values are bootstrap tail probabilities. Holm's procedure controls the family-wise error over HA, HD, HE and HF, with a non-estimable hypothesis counting as $p = 1$. Identical generations (same prompt text, guidance, steps and seed in one split) are counted once, two canonical scenes with

Table 3: Benchmark V2 design counts (scene graphs are the independent unit). TEST counts are design counts; the generated TEST shards are listed in Table 8.

Split	Scenes	Binding-bearing (literal)	Competitive subset	Spatial-bearing	Prompt variants
DEV	100	70	55	28	470
TEST	200	140	110	54	940

identical text count once, and test prompts whose text also occurs in the development split are excluded from variant-level summaries.

Sensitivity to judge error (exploratory, not registered). Restart persistence is a statement about agreement between judged outcomes, so judge errors enter both A and $A0$. If each judged outcome is flipped independently with probability e (symmetric, non-differential error), every observed agreement is the affine image $A^{\text{obs}} = (1-2e)^2 A + c$ of the true agreement with $c = 2e(1-e)$, for branch pairs and seed pairs alike. Substituting into the pooled estimator gives the error-corrected persistence

$$C_\kappa^{(e)}(t) = \frac{\sum_s \left[A_{\kappa s}^{\text{obs}}(t) - A0_{\kappa s}^{\text{obs}}\right]}{\sum_s \left[1 - c - A0_{\kappa s}^{\text{obs}}\right]}, \tag{7}$$

which equals the registered $C_\kappa(t)$ at $e = 0$. Error makes C shrink towards zero by an amount that depends on $A0$, which differs between guidance scales. The model assumes errors independent across images; a detector's errors are usually content-dependent (the same hard object is missed in every seed), which can move agreement in either direction and is not captured. We therefore use Eq. (7) as a stress test of how much judge noise a conclusion tolerates, not as a correction of the estimate, and evaluate $e \in \{0.05, 0.10\}$ and $e = 1 -$ (audited presence accuracy).

5 Benchmark

Prompts are generated from machine-readable scene graphs with a grammar and a lexicon. Every scene has a canonical prompt (CAN), meaning-preserving variants (synonym paraphrase PAR, mention-order change ORD) and counterfactual variants that change exactly one declared field (colour swap CF-C, relation reversal CF-R, count change CF-N, object deletion CF-O, attribute insertion or removal CF-A). Scenes are split into DEV and TEST at scene level, with object-pair groups confined to one split. The main benchmark has 100 DEV and 200 TEST scenes and 1,410 prompts (some variant types do not apply to some scene types). A factorial family of 24 object pairs (8 DEV, 16 TEST) crosses colour, relative size and spatial relation on top of presence, in the style of the task's example "a small red cube and a large blue sphere" (192 canonical prompts plus colour-swap, size-swap and relation-reversal counterfactuals).

Variant review and amendment. All 1,674 prompts and 73 lexicon entries were reviewed against the scene-graph constraints before generation. Two synonym choices were replaced in a dated amendment before any image of the affected prompts was generated: "golden" for yellow (the judge's text encoder placed it nearest to brown) and "on the left/right side of" (which admits the reference object's own frame); identifiers, constraints and splits were unchanged.

Table 4: DEV evaluator audit of J1 against blinded human labels (one annotator; population-weighted accuracy, stratified bootstrap 95% CI). Only types meeting ≥ 0.85 may carry a confirmatory claim.

Type	n	Accuracy	95% CI	≥ 0.85
presence	159	0.856	[0.802, 0.907]	yes
count	108	0.768	[0.685, 0.843]	no
binding	156	0.800	[0.722, 0.871]	no
spatial	109	0.615	[0.523, 0.706]	no
size	175	0.846	[0.794, 0.897]	no

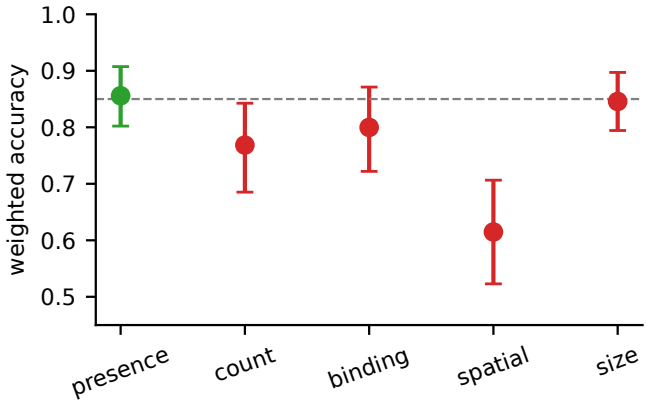


Figure 3: Weighted accuracy of the automatic judge per constraint type on DEV (95% bootstrap intervals; one annotator). Dashed line: pre-declared criterion 0.85. Green: met; red: not met.

6 Evaluation and judge audit

The judge (J1) combines open-vocabulary detection [20] with CLIP colour classification [24] at pinned model revisions, with thresholds fixed before any test data (presence 0.10, count 0.15, size ratio 1.5). A second, question-answering judge was planned but not implemented and plays no role in any result.

Pre-declared audit. A blinded, seeded, stratified sample of DEV (image, constraint) items was labelled by a human who saw only an opaque image and a neutral question (a colour, a count, a presence answer, which object is further left, or which looks larger), with "not visible" and "ambiguous" answers. Accuracy is population-weighted with a stratified bootstrap. A constraint type may carry a confirmatory claim only if its weighted accuracy is at least 0.85; no other criterion was added after the data were seen.

Result. Only presence met the criterion (Table 4, Figure 3). The binding judge reached 0.800, count 0.768, spatial relation 0.615 (its sensitivity was low) and relative size 0.846. One annotator labelled the main package; a second package was labelled by the same person within the same session and is therefore neither an inter-rater nor a valid intra-rater check, and is not used. These limits are carried into every result (Section 3).

Table 5: Frozen stage manifests: job counts and full-generation equivalents (40-step image cost) under the frozen constants.

Split	Stage	Content	Jobs	Equiv.
DEV	DEV_CORE	variants, CF bases, DEV size family	1,784	1,784
DEV	DEV_COMMIT	restarts D8/ER30/FULL, CFG 2/15 bases and restarts, switches	8,896	6,930
DEV	DEV_CAG_GUARD	CAG $w_{lo} \in \{1, 3\}$ quality guard	400	336
TEST	TEST_CORE	all variants; CFG × steps	9,160	7,660
TEST	TEST_COMMIT	restarts at CFG 2/7.5/15; prompt switches	10,560	6,986
TEST	TEST_CAG_SIZE	CAG arms A2/A3/A4/A6; size family with restarts and switches	5,536	4,923

7 Experimental design and execution

All jobs come from deterministic manifests sharded by scene group over three shards per stage (Table 5). Each stage ran only after the gates of the previous one had been checked. Test jobs could run only if the signed protocol verified against the code, configurations, benchmark, judge definition and development-set constants. Every output was merged against its manifest before analysis; missing, duplicated, corrupt or mismatching outputs are rejected, never silently dropped.

The development split comprises:

- the canonical DEV core: 400 images from 100 prompts and four seeds, with saved intermediate states; restarts branch from seeds 0 and 1;
- the DEV variants and size family (1,784 jobs);
- restart branches and CFG bases (8,896 jobs);
- the CAG quality guard (400 jobs).

The test split comprises 9,160 core and CFG×steps jobs, 10,560 restart and switch jobs, and 5,536 CAG and size-family jobs (19,569 full-generation equivalents). Six of its nine shards were generated: TEST_CORE shards 0–1, TEST_COMMIT shards 0–1 and TEST_CAG_SIZE shards 1–2 (Table 14). Generation used NVIDIA T4 GPUs at about 7 s per 40-step image.

8 Development-split results

Everything in this section uses DEV scenes only and was fixed before any test image was generated. It is exploratory; its role is to set constants and to decide, through pre-declared gates, which claims the test split can carry.

An estimator error found and corrected before the test split. The chance-agreement term $A0$ had first been computed from every core image of a scene, including paraphrases, counterfactual variants and other guidance scales, instead of only independent seeds of the branches' own base prompt as specified. This lowers $A0$ and biases every $C(t)$ upwards. The code was corrected, a regression test added, and the DEV constants were recomputed from the stored judgements; both versions are archived. All DEV numbers below use the corrected estimator.

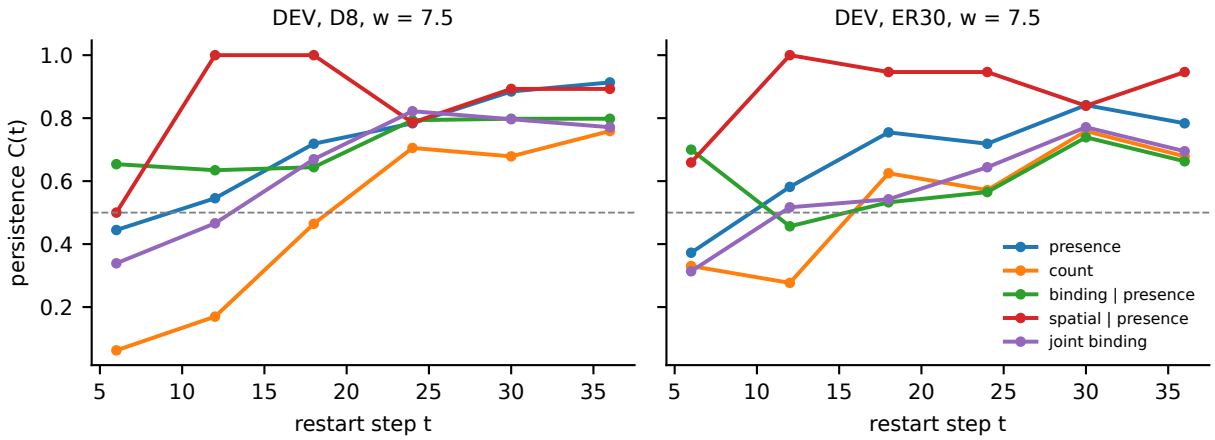


Figure 4: DEV restart persistence $C(t)$ at $w = 7.5$ for the fixed-jump (D8) and equal-ratio (ER30) probes (corrected estimator; exploratory). Dashed line: $C = 0.5$.

Table 6: DEV restart persistence t_{50} at $w = 7.5$ (steps of 40; exploratory). The analysis unit is the scene; restart branches exist for the 100 canonical DEV scenes whose seed-0 and seed-1 images were saved with intermediate states. †Left-censored: C already exceeds 0.5 at the first grid step.

Type	D8	ER30
presence	9.3	9.7
count	18.9	15.8
binding | presence	6.0†	6.0†
spatial | presence	6.0	6.0†
joint binding	13.0	11.5
Δt_{50} (binding − presence)	-3.29 [-7.56, 8.32]	-3.66 [-6.83, 6.77]
scenes with restart branches	100	100

Persistence curves. Figure 4 shows $C(t)$ on 100 DEV scenes. Presence crosses 0.5 near $t_{50} = 9.29$ steps (D8); colour binding given presence is already above 0.5 at the first grid step (left-censored, $t_{50} \leq 6.0$), and count is latest (18.89). With both restart probes the binding-minus-presence difference is negative but its interval includes zero: D8 $\Delta t_{50} = -3.29$ $[-7.56, 8.32]$, ER30 -3.66 $[-6.83, 6.77]$ (Table 6). Left-censoring means DEV cannot place binding relative to presence; and the binding judge missed the audit criterion.

Gates. Table 7 lists every pre-declared DEV gate. The HA precision gate failed (half-width 7.94 steps against a limit of 4), so HA is exploratory on the test split by the pre-declared rule. The full-resample control failed (maximum $|C| = 0.568$ against 0.15). A post-hoc diagnosis separates two components (Figure 13 in the appendix).

- *The gate statistic itself.* The maximum comes from spatial relation at $t = 6$, a cell informed by five scenes. Simulating exchangeable outcomes with the same design, scenes, eligibility pattern and frozen estimator, the gate statistic has a median of 1.0 and never fell below 0.15 in 1,000 simulations. As registered, the gate could not have been passed at this sample size, and its failure says nothing about bias.
- *A presence offset.* Presence, the best-supported cell, shows an offset of about +0.16 at

Table 7: DEV gates (corrected estimator). Failed gates are reported as failed; HA is therefore exploratory.

Gate	Criterion	DEV value	Result
HA precision	CI half-width ≤ 4 steps	7.94	failed
Full-resample control	$\max \lvert C \rvert \leq 0.15$	0.57	failed
HD presence, $w = 2$	≥ 40 scenes	45	passed
HD presence, $w = 15$	≥ 40 scenes	59	passed
CAG quality (HPS v2), $w_{lo} = 1$	mean drop ≤ 0.5	-0.52	failed
CAG quality (HPS v2), $w_{lo} = 3$	mean drop ≤ 0.5	-0.26	passed

every t, beyond the simulated null (simulation $p = 0.008$). The estimator is unbiased under exchangeability in those simulations and in a unit test, and a FULL branch is a genuinely fresh draw. Its noise seed is hashed per job and never coincides with a base seed, and the marginal presence rates of FULL branches and base seeds are equal (0.389 against 0.390). What differs is agreement: FULL branches agree with one another more often (0.73) than the four base seeds do (0.65; difference 0.076, CI $[0.028, 0.126]$). FULL images also agree with each base seed, including the two seeds that were never restarted, more often (about 0.70) than the seeds agree among themselves.

Two explanations remain, and we cannot separate them. First, every prompt uses the same four initial-noise seeds, so A_0 is measured on one fixed panel of four noise tensors shared by all scenes, and its error is common to all scenes; a seed main effect is not detectable (permutation $p = 0.36$). Second, there may be an undetected difference between the base images, generated in an earlier run, and the branch images. A loop-identity check passed on three prompts but not on these images. The offset would make $C(t)$ uniformly too high and absolute t_{50} too early. It affects contrasts less: subtracting it at every guidance scale leaves the HD contrast intact (Section 10). The smallest experiment that would decide between the two explanations is about 400 images: regenerate FULL branches for 50 DEV scenes with the panel seeds 2 and 3 through the branch code (any outcome difference from the stored seed images exposes a pipeline difference), and draw four further base seeds per scene to compare A_0 across panels. Both HD presence gates passed (45 and 59 scenes).

CAG quality guard. HPS v2 scored the 200 DEV guard images per w_{lo} against their constant-CFG partners (same prompt and seed). With $w_{lo} = 1$ the mean change was -0.518 HPS points (95% interval -0.622 to -0.418), a drop just beyond the pre-declared 0.5; with $w_{lo} = 3$ it was -0.259 (-0.324 to -0.204). By the pre-declared rule $w_{lo} = 3$ was adopted before the test split (Figure 5). The switch steps derived from DEV are $s^\star = 12$ for prompts whose latest type is presence, binding or spatial relation and $s^\star = 21$ for prompts with counts (global $s^\star = 12$).

9 Registered analysis on the partial test split

This section applies the registered estimators and decision rules, unchanged, to the six test shards that exist. Because three shards are missing, every result is **partial and exploratory**: it covers a subset of the test scenes, and the Holm-adjusted p-values are reported over the full registered family for completeness, not as a confirmatory test (Table 9). Only presence has a validated judge (Section 6). HF's outcome uses the registered rejection rule (concordance ≤ 0.5, Section 3).

Limitations of this analysis.

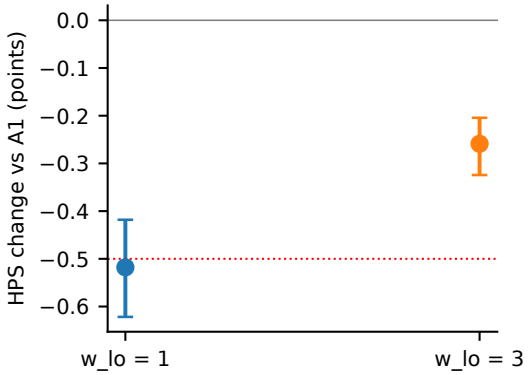


Figure 5: DEV CAG quality guard: mean HPS v2 change of CAG images relative to constant CFG 7.5 (95% scene-bootstrap intervals). Dotted line: pre-declared limit −0.5.

Table 8: TEST coverage of this PARTIAL TEST, EXPLORATORY analysis. Shards are disjoint scene groups; missing shards are not imputed.

Stage	Shards present	Missing	Jobs merged	Scenes
TEST_CORE	[0, 1]	[2]	6172/6172	133
TEST_COMMIT	[0, 1]	[2]	7080/7080	85
TEST_CAG_SIZE	[1, 2]	[0]	3750/3750	222

- Missing TEST shards (not imputed): TEST_CORE shard 2; TEST_COMMIT shard 2; TEST_CAG_SIZE shard 0; estimates cover only the scene groups of the present shards.
- This is not the registered confirmatory analysis, which requires all nine shards; Holm-adjusted p-values are shown for completeness only.
- HA is exploratory by its failed DEV precision gate; the full-resample control failed on DEV.
- Only the presence judge met the audit criterion (single annotator, no valid double annotation); count, binding, spatial and size outcomes are exploratory.
- HE contrasts with constant guidance use only the 67 scenes with both A1 and CAG images (one CAG shard); the comparison with the reversed schedule uses all scenes with CAG arms.

10 Post-hoc analyses of the partial test split

The analyses below were chosen after the registered outcomes of Section 9 were known. They are exploratory, uncorrected for multiplicity, and computed from the saved judgements with the frozen estimator code; no image was generated or re-judged. Table 14 defines every sample size used. Where a number duplicates a registered output, the registered value is reported.

Integrity of the partial evidence. Every image of the six present shards was judged exactly once and has one DINOv2 embedding. After the registered de-duplication, re-judging all 16,974 images on different hardware and software reproduced every judged outcome; all four rule outcomes were unchanged and bootstrap p-values differed by at most 0.002.

Table 9: Hypotheses on the available TEST shards (PARTIAL TEST, EXPLORATORY). p_{Holm} over the registered family is shown for completeness only; no result here is a confirmatory finding. p is the registered hypothesis-level p (for HD, $\min(1, 2 \min p)$ over its two estimands; Table 11).

Hyp.	Estimand	Estimate	CI	p	p_{Holm}	Rule outcome on partial data
HA	Δt_{50} binding−presence (steps)	3.59	[-10.43, 13.06]	0.5652	1.0000	EXPLORATORY (DEV precision gate failed (half-width 7.94 > 4); binding judge below the audit criterion); rule outcome: inconclusive
HD	Δt_{50} presence, $w15{-}2$ (steps)	-8.67	[-16.34, -4.72]	0.0116	0.0464	supported
HE	A2−A1 on F (90% CI; margin −0.03)	-0.016	[-0.035, 0.003]	0.4825	1.0000	not supported (non_inferior, beats_reversed failed)
HF	type-order concordance	0.30	[0.20, 0.75]	0.8042	1.0000	rejected (ordering concordance ≤ 0.5)

Table 10: Persistence steps t_{50} at $w = 7.5$ on TEST shards present (PARTIAL TEST, EXPLORATORY; HA is exploratory by its DEV gate). †left-censored.

Kind	D8	ER30
presence	16.2	13.3
count	15.3	14.0
binding | presence	19.8	21.0
spatial | presence	6.0†	6.0†
size | presence	13.6	–
joint binding	21.2	21.3
Δt_{50} (binding−presence) [95% CI]	3.59 [-10.43, 13.06]	7.67 [-12.64, 24.88]
scenes	111	67

What guidance changes about timing (HD). Presence persistence comes earlier as guidance increases: $t_{50} = 22.5$, 16.2 and 13.8 steps at $w = 2$, 7.5 and 15 (Table 15, Figure 7a). The ordering is not manufactured by the chance correction. Chance agreement is not higher at $w = 15$ ($\bar{A}_0 = 0.697$ against 0.729 at $w = 2$), and uncorrected branch–base agreement at $t = 18$ is already higher at $w = 15$ (0.90 against 0.79). Table 16 reports four further checks:

- Bootstrapping only the 67 scenes that carry both guidance levels gives practically the same interval.
- Subtracting the full-resample offset found on DEV (Section 8) from $C(t)$ at every w leaves the contrast at -10.96 steps, CI $[-17.95, -3.83]$.
- The sign holds in both TEST_CORE shards and in four of five scene categories, but reverses in the 10 counting scenes (+11.4 steps).
- Under the independent-error stress test of Eq. (7), the interval reaches zero at $e = 0.05$ and $e = 0.10$. At the audited presence error rate ($e = 0.144$), every corrected curve exceeds 0.5 at the first grid step, so the contrast is no longer identifiable (Figure 7b).

The HD result therefore survives the estimator-level threats we can test, but it rests on the presence judge's errors being largely content-dependent rather than independent across seeds. The audit did not measure that.

Table 11: HD on the available shards (PARTIAL TEST, EXPLORATORY): t_{50} by guidance scale and the per-estimand bootstrap p. The registered HD p in Table 9 is $\min(1, 2 \min p)$ over the two estimands. Binding is exploratory by the audit; n.e.: not estimable because binding never reaches $C = 0.5$ at $w = 2$ (right-censored).

Estimand	t_{50} (w=2 / 7.5 / 15)	Δt_{50} (15−2)	95% CI	p	rule
presence	22.5 / 16.2 / 13.8	-8.67	[-16.34, -4.72]	0.0058	supported
binding | presence	> 36 / 19.8 / 12.5	n.e.	[-24.00, 1.50]	0.0742	inconclusive

Table 12: HE: CAG ($w_{lo} = 3$) vs constant CFG 7.5 (A1) and the reversed schedule A4 (PARTIAL TEST, EXPLORATORY). Contrasts with A1 use the 67 scenes that have both A1 and CAG images; A2−A4 uses every scene with CAG arms (Table 14).

Contrast	Estimate	CI	Requirement	Met
A2−A1 on F	-0.016	[-0.035, 0.003]	90% CI lower > -0.03	no
A2−A1 diversity	0.0219	[0.0151, 0.0286]	95% CI > 0	yes
A2−A4 on F	0.001	[-0.025, 0.027]	95% CI > 0	no
A3−A1 on F	-0.031	[-0.056, -0.006]	secondary	–
A6−A1 on F	-0.005	[-0.041, 0.029]	secondary	–

What guidance changes about the images. On the 102 scenes generated at every guidance scale and step count (Figure 8), raising w from 2 to 15 increases the presence rate (0.32 to 0.43) and whole-prompt satisfaction (0.09 to 0.17), with little change beyond $w \approx 10$. Over the same range the CLIP score rises from 29.9 to 32.5, and seed diversity falls from 0.668 to 0.478. Doubling the number of DDIM steps from 20 to 40 changes neither presence nor whole-prompt satisfaction at any w (at $w = 7.5$: -0.002, CI $[-0.034, 0.029]$; Figure 14a in the appendix). Adherence falls steeply with prompt complexity: whole-prompt satisfaction is 0.33 for prompts with 2 constraints and at most 0.07 for prompts with four or more (Figure 14b). These are associations across the guidance grid on canonical prompts, not a mechanism.

Commitment times do not transfer from DEV to TEST. DEV placed colour binding at or before the first grid step, presence at 9.3 and count last (18.9). The partial test split places count (15.3) and presence (16.2) before binding (19.8), and joint binding moves from 13.0 to 21.2 (Table 17). Both restart probes show the shift, and the DEV presence value lies outside the TEST interval. Several causes are possible and cannot be separated here:

- the TEST estimate covers two of three shards;
- DEV failed its own precision gate;
- the binding judge is not validated;
- the DEV binding curve is censored at the first grid step.

The shift matters for CAG. Its switch step for presence, binding and spatial prompts is $s^\star = \lceil \max(9.3, 6.0) \rceil + 2 = 12$, where the binding value of 6.0 is DEV's left-censored t_{50} entering the frozen rule as a number. On the test scenes binding is not yet persistent at step 12.

Table 13: HF: attention stabilisation vs persistence (PARTIAL TEST, EXPLORATORY).

Quantity	Value
concordance	0.30
95% CI	[0.20, 0.75]
pairs	10
Spearman ρ, binding | presence	0.20 [-0.37, 0.69]
Spearman ρ, count	-0.30 [-0.83, 0.43]
Spearman ρ, presence	-0.19 [-0.42, 0.07]
Spearman ρ, size | presence	0.12 [-1.00, 1.00]
Spearman ρ, spatial | presence	0.37 [-0.43, 0.89]
rule	rejected (ordering concordance ≤ 0.5)

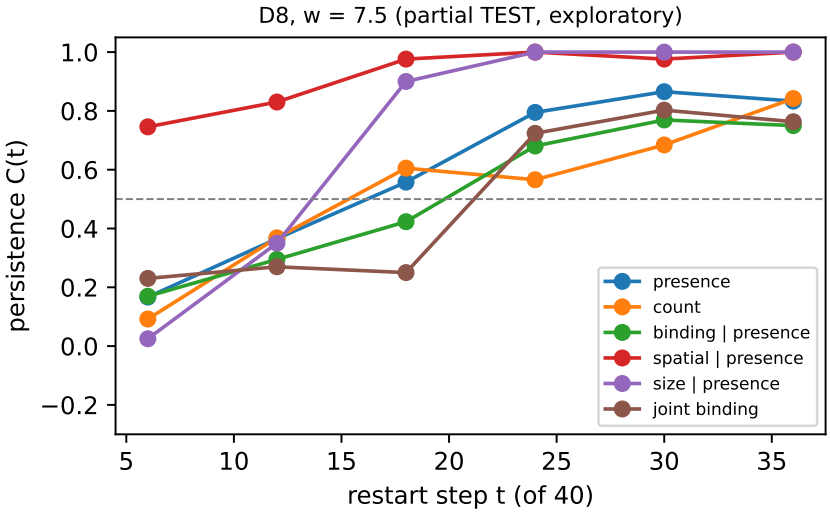


Figure 6: Chance-corrected persistence $C(t)$ by constraint type under the D8 restart probe ($w = 7.5$); dashed line marks $C = 0.5$, whose crossing defines t_{50}. Partial test split, exploratory.

CAG: dose, not timing (HE). The registered HE contrast compares CAG with constant $w = 7.5$, but every CAG arm applies less guidance on average ($\bar{w} = 4.64$ for A2, A4 and A6). The matched-mean controls separate the two factors (Table 18, Figure 9):

- The registered diversity gain of A2 over A1 (+0.022) is matched by constant guidance at the same mean (A6, +0.027) and exceeded by the reversed schedule (A4, +0.040).
- At equal mean guidance, A2 is less diverse than A6 (-0.013, CI $[-0.026, -0.001]$) and than A4 (-0.030), and its adherence does not differ from A6 ($\Delta F = -0.002$, CI $[-0.026, 0.023]$).
- The single global switch step (A3) lowers adherence (registered secondary contrast -0.031, CI $[-0.056, -0.006]$).

On these data, concentrating guidance before the DEV commitment step buys nothing that the same amount of constant guidance does not. It also costs diversity relative to spending the guidance late. This agrees with reports that guidance at high noise levels reduces diversity [16] and that late conditioning improves alignment [8].

Table 14: What the partial TEST split contains and which scenes enter each estimate (scene = independent unit; seeds, branches and arms of a scene are resampled together). Partial TEST.

Stage or estimate	Units	Composition
TEST_CORE (shards 0, 1)	133 scene IDs	canonical, variant and CFG × steps images; attention maps
TEST_COMMIT (shards 0, 1)	85 scene IDs	67 with restart branches, 54 with prompt-switch jobs
TEST_CAG_SIZE (shards 1, 2)	222 scene IDs	134 main-benchmark scenes with CAG arms; 88 size-family scenes (11 object-pair groups), 44 of them with restart branches
HA, D8 ($w = 7.5$)	111 scenes	67 TEST_COMMIT + 44 size-family scenes
HA, ER30	67 scenes	TEST_COMMIT only
HD	67 scenes	branches at $w = 2$ and 15 (TEST_COMMIT); the registered bootstrap resamples all 111
HE, A2 vs A1 (adherence, diversity)	67 scenes	A1 needs a TEST_CORE image; only CAG shard 1 overlaps
HE, A2 vs A4; dose-matched contrasts	134 scenes	all scenes with CAG arms
HF	221 scenes	2,516 attention maps at $w = 7.5$

The registered HE output lists 288 scenes: every scene with an adherence value in A1 or any CAG arm, not the number of paired scenes.

Table 15: Commitment time by guidance scale on canonical prompts (restart D8, 40 steps; partial TEST, exploratory). $\bar{A}_0$: mean chance agreement of independent seeds.

w	t_{50} presence	95% CI	$\bar{A}_0$ presence	t_{50} binding|pres.
2	22.5	[19.5, 27.9]	0.729	–
7.5	16.2	[12.4, 19.1]	0.649	19.8
15	13.8	[6.9, 17.2]	0.697	12.5

"–" marks a curve that never reaches $C = 0.5$ on the grid (right-censored, $t_{50} > 36$).

11 What cross-attention shows (post hoc)

The test generations saved, for every canonical image, the cross-attention map of each constraint word at all 40 steps. That gives 2,516 maps on 221 scenes: guidance $w \in \{2, 5, 7.5, 10, 15\}$ for the main benchmark and $w = 7.5$ for the size family, all at 40 steps. No maps exist for 20-step images, restart branches, prompt switches or the CAG arms, so attention cannot be compared across schedules. All analyses below are post hoc. They use 16×16 maps, each word's map normalised to unit mass at every step. *Localisation* is the share of a noun's mass in its top 10% of cells, and *overlap* is the histogram intersection with the other object nouns of the prompt.

Attention separates found from missed objects from the first steps. At $w = 7.5$ we compare, within the same scene and object, seeds in which the judge found the object with seeds in which it did not (226 scene–object pairs in 159 scenes; Figure 10a,b). The noun's attention is more localised and overlaps less with the other nouns when the object is later found. Both differences are distinguishable from zero from the first step; at step 10 they are +0.062 (CI $[0.049, 0.075]$) and -0.051 ($[-0.066, -0.037]$). This is a within-scene association, not a causal effect: the initial noise fixes both the early attention layout and, often, the outcome. It is consistent with object neglect being visible early in attention [3] and with forecasting final quality from early attention

Table 16: Robustness of the HD presence contrast $\Delta t_{50} = t_{50}(w{=}15) - t_{50}(w{=}2)$ (steps; scene-bootstrap 95% CI). Judge-error rows correct $C(t)$ for symmetric, non-differential misclassification at rate e (Eq. 7); the offset row subtracts the presence offset of the DEV full-resample control from $C(t)$ at every w. All rows use the registered bootstrap draws; "scenes" counts scenes with branches at both $w = 2$ and $w = 15$. Partial TEST, exploratory.

Analysis	scenes	Δt_{50}	95% CI
registered estimate and interval ($e = 0$; bootstrap over all 111 scenes)	67	-8.67	[-16.34, -4.72]
resampling only the scenes with $w = 2$ and $w = 15$ branches	67	-8.67	[-16.25, -4.63]
judge error $e = 0.050$	67	-12.91	[-15.46, 0.71]
judge error $e = 0.100$	67	-5.55	[-13.41, 0.00]
judge error $e = 0.144$ (audit)	67	0.00	[-8.46, 0.00]
$C(t) - 0.159$ (DEV full-resample presence offset)	67	-10.96	[-17.95, -3.83]
TEST_CORE shard 0 only	33	-6.64	[-15.52, 5.60]
TEST_CORE shard 1 only	34	-9.66	[-22.00, -5.65]
category: binding	20	-16.44	[-24.00, -4.91]
category: counting	10	11.40	[9.00, 15.00]
category: multi	17	-7.71	[-18.50, 9.67]
category: presence	10	-10.00	[-21.00, 10.67]
category: spatial	10	-0.67	[-12.40, 22.33]

Subgroup rows have few scenes; an interval that includes 0 there is a statement about power, not about the effect.

[4]. Figure 15 in the appendix shows three pairs chosen by a fixed rule (the three lowest scene identifiers with a two-object prompt and both outcomes; the lowest-index seed of each outcome).

Guidance separates the nouns. On identical initial noise (same scene, seed and object), higher guidance lowers the overlap between object nouns throughout the trajectory. At step 10 overlap falls from 0.688 at $w = 2$ to 0.623 at $w = 15$; the paired difference is -0.065, CI $[-0.081, -0.051]$ (Figure 10c). Over the same range the share of objects found rises from 0.57 to 0.64. Figure 11 shows one scene, chosen by rule as the lowest-identifier binding scene with maps at all three scales. At $w = 2$ the two nouns' maps stay mixed and the bottle is not drawn; at $w = 7.5$ and 15 they separate within the first ten steps.

Why attention stabilisation does not order the constraint types (HF). The registered statistic, the step t_{stab} at which a type's attention signal has travelled half of its total path, does not order types the way restart persistence does (Figure 12a).

- At $w = 7.5$, t_{stab} spans only 8.2–14.7 steps across types, against 6–20 for t_{50}.
- Binding is the type with the earliest stabilisation (8.2) and the latest persistence (19.8).
- The binding signal is the colour word's attention map, which largely follows the object's location. Its stabilisation (8.2) is close to that of presence (9.3) at every guidance scale. The signal therefore records where the colour will be placed, not whether the colour–object assignment has been made.

Attention does, however, respond to guidance in the same direction as persistence. Between $w = 2$ and $w = 15$, t_{stab} moves earlier by -9.6 steps for presence (paired CI $[-10.0, -9.2]$) and by -9.1 for binding, but only by -2.7 for count (Figure 12b). For presence this is the same direction and a similar size as the HD restart contrast. The two signals agree on *what guidance does*, yet disagree

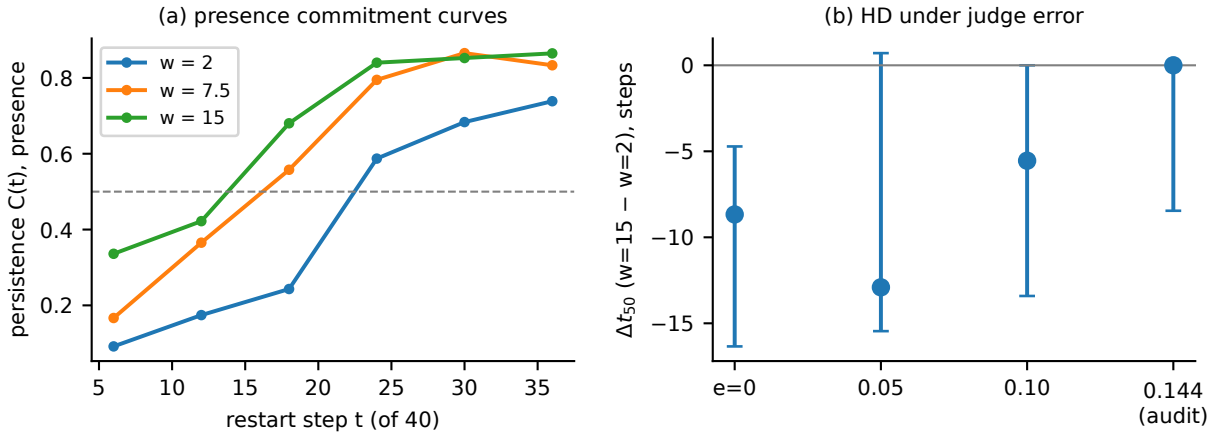


Figure 7: **Partial test split, post hoc.** (a) Presence persistence $C(t)$ at three guidance scales (restart D8, canonical prompts, 67 scenes). (b) The HD contrast $t_{50}(w{=}15) - t_{50}(w{=}2)$ with 95% scene-bootstrap intervals: registered estimate ($e = 0$) and the independent-error stress test of Eq. (7). At the audited error rate all curves are left-censored and the contrast is 0 by construction.

Table 17: Commitment time t_{50} (steps of 40, $w = 7.5$) on DEV (the values that set $s^\star$) and on the partial TEST split. ≤ 6: left-censored at the first grid step. Only presence has a validated judge. Exploratory.

Type	DEV D8	TEST D8	DEV ER30	TEST ER30
presence	9.3	16.2	9.7	13.3
count	18.9	15.3	15.8	14.0
binding|presence	≤ 6	19.8	≤ 6	21.0
spatial|presence	6.0	≤ 6	≤ 6	≤ 6
size|presence	–	13.6	–	–
joint binding	13.0	21.2	11.5	21.3
scenes	100	111	100	67

DEV sets the frozen switch steps $s^\star = 12$ (presence, binding, spatial prompts) and $s^\star = 21$ (count prompts).

on *which constraint settles first*. This is the pattern expected if attention tracks spatial layout rather than attribute binding.

12 Registered confirmatory analysis

PENDING (requires all nine test shards). The registered confirmatory test has not been run, because three of nine test shards were not generated:

- TEST_CORE shard 2 holds a third of the base images and attention maps, which every hypothesis needs.
- TEST_COMMIT shard 2 holds a third of the restart branches used by HA and HD.
- TEST_CAG_SIZE shard 0 holds a third of the CAG arms and size-family scenes used by HE and by HA's size estimates.

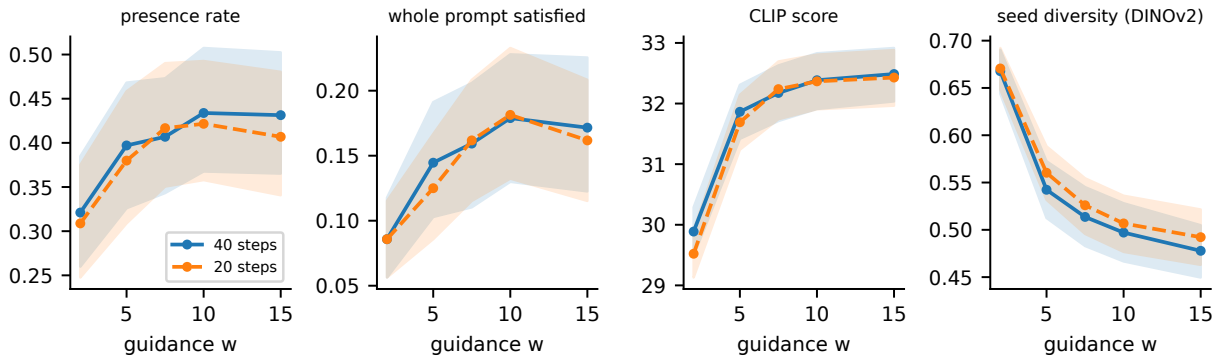


Figure 8: **Partial test split, post hoc.** Canonical prompts on 102 scenes common to every cell: presence rate, whole-prompt satisfaction, CLIP score and seed diversity (mean pairwise DINOv2 cosine distance over the four seeds) against the guidance scale, at 40 and 20 DDIM steps (scene means, 95% scene-bootstrap bands).

Table 18: Guidance-schedule contrasts paired by scene (difference, 95% CI); A1 is constant $w = 7.5$. $\bar{w}$: mean applied guidance of the two arms. Presence is the only judged family that met the audit criterion. Partial TEST, exploratory.

Contrast	$\bar{w}$	scenes	ΔF	Δ presence	Δ diversity
A2 (CAG) − A1	4.64 vs 7.50	67	-0.016 [-0.039, 0.006]	-0.026 [-0.052, 0.000]	0.022 [0.015, 0.029]†
A3 (global $s^\star$) − A1	4.35 vs 7.50	67	-0.031 [-0.056, -0.006]†	-0.045 [-0.082, -0.015]	0.030 [0.022, 0.038]
A4 (reversed) − A1	4.64 vs 7.50	67	-0.021 [-0.062, 0.017]	-0.015 [-0.063, 0.034]	0.040 [0.016, 0.064]
A6 (constant at A2 mean) − A1	4.64 vs 7.50	67	-0.005 [-0.041, 0.029]†	-0.015 [-0.063, 0.030]	0.027 [0.009, 0.045]
A2 − A6 (timing at equal dose)	4.64 vs 4.64	134	-0.002 [-0.026, 0.023]	-0.011 [-0.043, 0.021]	-0.013 [-0.026, -0.001]
A2 − A4 (registered timing contrast)	4.64 vs 4.64	134	0.001 [-0.025, 0.027]†	-0.015 [-0.049, 0.021]	-0.030 [-0.044, -0.015]

†Output of the registered analysis (its own bootstrap draws); other cells are post-hoc. "scenes": scenes with both arms. A1 exists only for scenes of the present TEST_CORE shards, so contrasts with A1 use 67 scenes (all in CAG shard 1); contrasts between CAG arms use all 134.

Its absence also limits HE's comparison against constant guidance to the 67 scenes of one CAG shard (Table 14). Once the shards exist, the frozen pipeline will replace this section with the registered result, without any change to estimators, rules or constants. No statement in this paper is a confirmatory finding.

13 Discussion

What CFG changes about timing, and how sure we are. Across three guidance scales, object presence becomes persistent earlier under stronger guidance. Under a restart that re-noises by eight steps, a trajectory at $w = 15$ has settled whether its objects appear several steps before one at $w = 2$. Three observations make this more than a statistical accident of one estimator:

- the ordering holds in uncorrected agreement;
- it survives subtracting the unexplained offset of the full-resample control;
- cross-attention shows a parallel shift on identical initial noise: object nouns separate earlier and their maps stabilise about ten steps earlier.

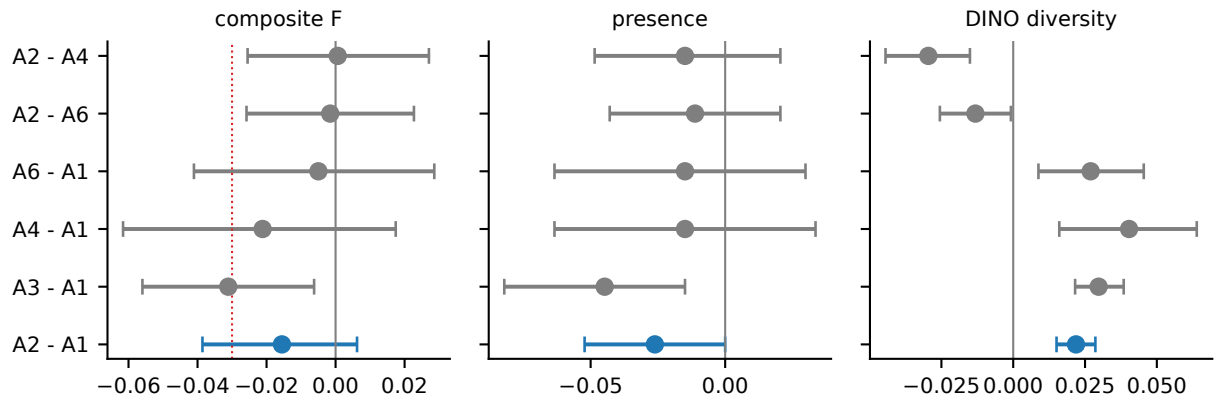


Figure 9: **Partial test split, post hoc.** Paired scene-level contrasts of the guidance schedules (95% intervals). Rows "Ak − A1" compare against constant $w = 7.5$ (67 scenes); "A2 − A6" and "A2 − A4" compare schedules with equal mean guidance (134 scenes). Dotted line: the registered non-inferiority margin −0.03 on F.

Three things limit it.

- Only three guidance levels were measured, so we cannot describe the shape of the relation.
- The contrast rests on the 67 scenes with branches at both extremes and reverses in the counting scenes.
- Correcting for a judge that errs independently on 14% of images, the audited rate, removes it.

Our audit measured how often the presence judge errs, not whether its errors repeat across the seeds of a scene, and that is the property the conclusion depends on. A seed-paired audit, labelling several seeds of the same prompt, is the cheapest way to settle it.

Attention shows where things will be, not what has been bound. The saved attention maps add two things the restart probe cannot. First, objects that the judge later misses differ from found objects of the same scene from the first denoising steps: their noun attention is less localised and overlaps more with the other nouns. Object neglect is therefore visible long before the image is decided, consistent with early-attention forecasting [4, 9] and with the premise of attention-based corrections [3]. Second, the registered attention-stabilisation statistic fails as a stand-in for persistence across constraint types, for a specific reason. Its binding signal is the colour word's map, which follows the object's location and so stabilises with the layout, around step 8 at $w = 7.5$. Restart persistence of the binding itself comes about twelve steps later. Attention stabilisation is a reasonable proxy for when the layout settles, and a poor one for when an attribute is bound.

Commitment times are not yet a stable property. The development split suggested that colour assignment is fixed at least as early as object presence and that numerosity stays open longest. The test scenes do not reproduce this: binding becomes persistent after presence under both restart probes. Several causes could produce the shift and the partial data cannot separate them:

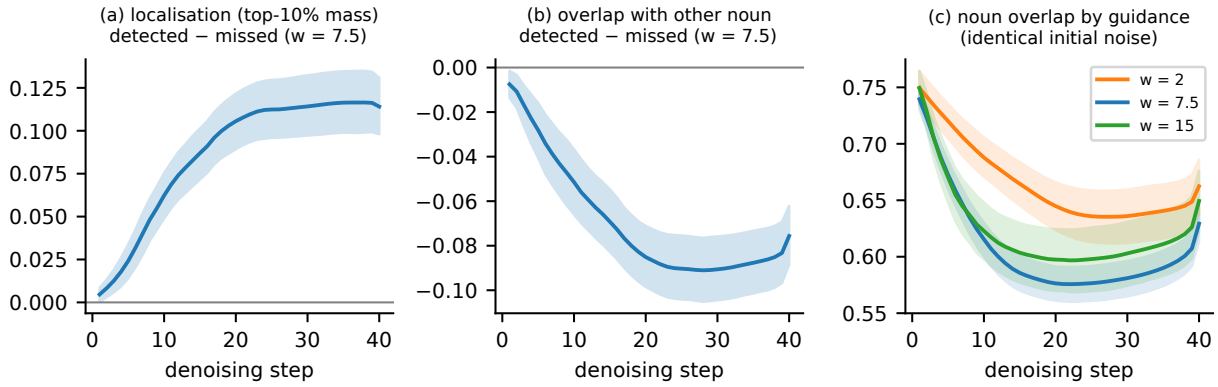


Figure 10: **Partial test split, post hoc.** (a,b) Within-scene difference, found minus missed, in a noun's attention localisation and in its overlap with the other nouns at $w = 7.5$ (226 scene–object pairs, 159 scenes). (c) Noun overlap at three guidance scales on identical initial noise. Bands: 95% scene-bootstrap intervals.

- a different scene composition;
- the DEV precision gate that had already failed;
- the unvalidated binding judge;
- left-censoring of the DEV binding curve.

We draw one methodological conclusion: a commitment time estimated on about 100 scenes, with intervals wider than ± 7 steps, is too imprecise to schedule an intervention on.

Why CAG did not help. CAG was meant to spend guidance while a constraint is still open and release it afterwards. It failed for two identifiable reasons. Its switch steps came from DEV commitment times that did not transfer, so on the test scenes it lowers guidance at step 12 while binding is still unsettled. And its registered comparison confounds timing with dose: every CAG arm applies less mean guidance than constant $w = 7.5$, and the matched-mean controls attribute the diversity gain to that lower mean. At equal mean guidance, front-loading is the least diverse of the three orders and no better in adherence. This agrees with interval guidance [16], where guidance at the highest noise levels harms diversity, and with step selection [8], where late conditioning helps alignment more than early conditioning. The matched-mean design is what made this diagnosis possible, and we recommend it for any guidance-schedule claim. Online, prompt-dependent schedules [22] avoid the transfer problem by measuring during sampling and are the natural comparison for a revised CAG.

The estimator's open validity question. The full-resample control was meant to show $C \approx 0$ for independent draws. Its registered gate could not pass at this sample size, which we only recognised after the fact. Its presence offset is real, though, and unexplained: either the fixed four-seed noise panel makes A_0 unrepresentative, or base and branch images differ in a way the identity checks did not cover. Both explanations bias absolute t_{50} more than contrasts between conditions. A 400-image experiment would separate them (Section 8). Until then, absolute commitment steps in this paper should be read as probe- and panel-relative.

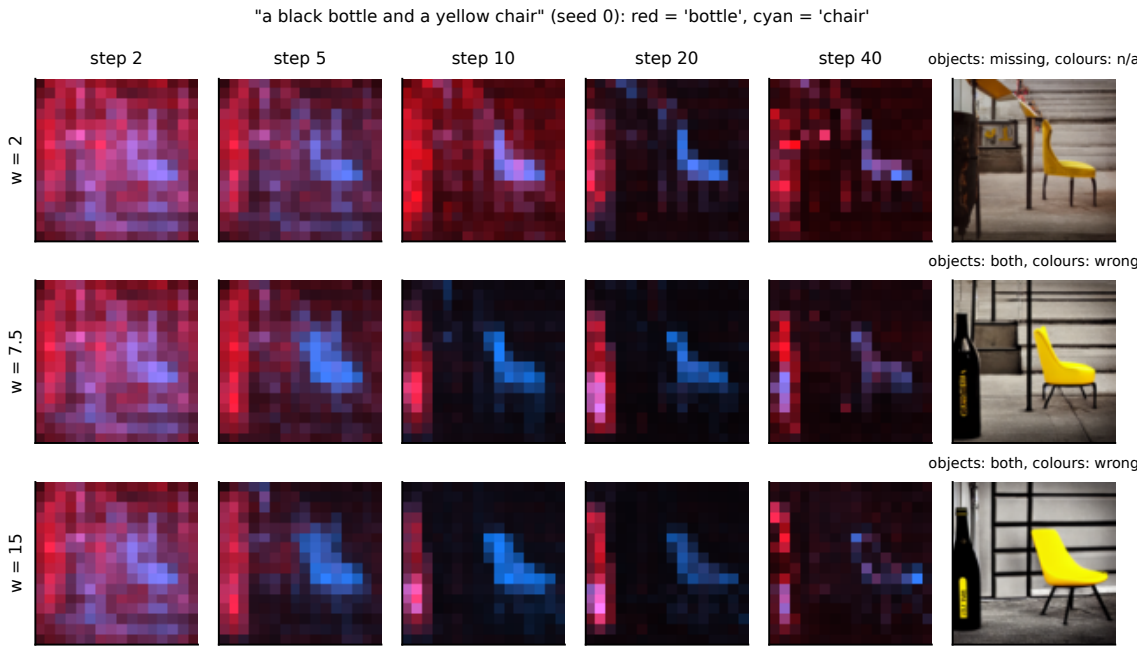


Figure 11: **Illustration, not evidence.** Attention of the two object nouns (red, cyan) at five denoising steps for one prompt and seed at three guidance scales (same initial noise), with the judged outcome. Selection rule: the lowest-identifier binding scene whose seed-0 maps exist at $w = 2$, 7.5 and 15.

14 Limitations and threats to validity

1. **Incomplete test split.** Three of nine test shards are missing. All test results are partial and exploratory, and the registered confirmatory analysis has not been run (Section 12).

2. **Judge validity and error structure.** Only presence met the pre-declared accuracy criterion; results for binding, count, spatial relation and relative size are exploratory. One annotator labelled the audit, and no valid double annotation exists. The HD conclusion depends on how presence errors are distributed across seeds, which the audit did not measure.

3. **Estimator validity.** The full-resample control shows an unexplained positive presence offset. Absolute t_{50} values are therefore probe- and seed-panel-relative. Contrasts between guidance scales are less affected (Section 10). The registered gate for this control could not have been passed at the DEV sample size.

4. **Transfer of DEV constants.** Commitment times did not replicate from DEV to the test scenes, so CAG was tested with switch steps that are mis-timed for binding.

5. **Dose confound in the registered HE contrast.** CAG is registered against constant $w = 7.5$ at a lower mean guidance. Only the post-hoc matched-mean contrasts separate timing from dose.

6. **Post-hoc analyses.** Sections 10 and 11 were chosen after the registered outcomes were known and are not corrected for multiplicity. Subgroups rest on 10–34 scenes.

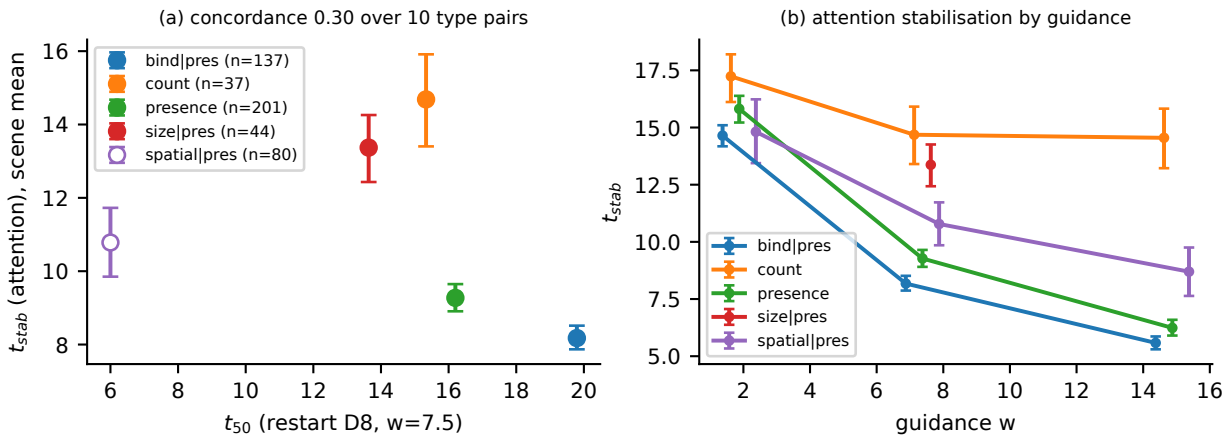


Figure 12: **Partial test split; (a) registered quantities, (b) post hoc.** (a) Attention stabilisation t_{stab} (scene means, 95% intervals) against restart persistence t_{50} per type at $w = 7.5$; open marker: left-censored t_{50}. (b) t_{stab} per type at three guidance scales.

7. **Units and dependence.** Scenes are resampled as independent units. The 88 size-family scenes come from 11 object pairs, so their independence is overstated. The registered HD bootstrap resamples 111 scenes, of which 67 inform the contrast.

8. **Attention coverage.** Maps were saved only for canonical 40-step images: none for restart branches, prompt switches, 20-step images or CAG arms. Attention results are associations, not interventions.

9. **Probe dependence and censoring.** Restart persistence is defined relative to a re-noising magnitude, and several t_{50} values are censored at the first grid step.

10. **Selection.** Conditional outcomes (binding, spatial relation and size given presence) condition on a generated variable.

11. **Scope.** One model (Stable Diffusion 1.5), one sampler and resolution, and synthetic prompts; some colour–object pairs are unnatural by design.

12. **Literature check.** Most related works were checked at abstract level; novelty is claimed only for the specific combination stated in Section 2.

13. **AI assistance.** Code, analysis scripts and manuscript drafts were produced with AI assistance under the author's direction. The DEV audit labels were produced by the author without AI assistance; the pilot validation labels were drafted with AI assistance and corrected by the author (Appendix B).

15 Reproducibility and provenance

All decision rules, development-set constants and test manifests were frozen in a signed protocol before any test image was generated. Every reported number is generated from saved, checksummed outputs by versioned scripts, and generation itself is deterministic given the manifest, so any result can be recomputed without regenerating images. Appendix C lists the generators and

the provenance record. Deviations from the registered plan are disclosed where they occur: the chance-agreement correction before the test split (Section 8), the single-annotator audit (Section 6), and every post-hoc analysis (Sections 10 and 11).

16 Conclusion

We asked when compositional constraints become persistent during diffusion sampling and what classifier-free guidance changes about that timing. On the available two-thirds of a pre-registered test split, stronger guidance makes object presence persistent earlier, and cross-attention shows the same shift: object nouns separate earlier on identical noise, and missed objects are visible as diffuse, overlapping attention from the first steps. Both findings are exploratory. The timing result holds only if the presence judge's errors are not independent across seeds.

The intervention built on the measurement did not work. Commitment times from the development split did not transfer, and Commitment-Aligned Guidance neither kept adherence nor beat its reversed schedule; matched-mean controls showed that its extra diversity came from lower average guidance. The broader lesson is about precision. Guiding by commitment time requires timing estimates far tighter than about 100 scenes provide, and judges that are validated, including for their error structure, for more than object presence. Three steps remain:

- the registered confirmatory analysis, once the three missing shards exist;
- a seed-paired judge audit;
- a 400-image experiment that resolves the full-resample offset.

References

[1] Yogesh Balaji, Seungjun Nah, Xun Huang, Arash Vahdat, Jiaming Song, Qinsheng Zhang, Karsten Kreis, Miika Aittala, Timo Aila, Samuli Laine, Bryan Catanzaro, Tero Karras, and Ming-Yu Liu. eDiff-I: Text-to-image diffusion models with an ensemble of expert denoisers, 2022. arXiv:2211.01324.

[2] Yuanhao Ban, Ruochen Wang, Tianyi Zhou, Minhao Cheng, Boqing Gong, and Cho-Jui Hsieh. Understanding the impact of negative prompts: When and how do they take effect?, 2024. arXiv:2406.02965.

[3] Hila Chefer, Yuval Alaluf, Yael Vinker, Lior Wolf, and Daniel Cohen-Or. Attend-and-excite: Attention-based semantic guidance for text-to-image diffusion models. *ACM Transactions on Graphics*, 42(4), 2023. arXiv:2301.13826.

[4] Benlei Cui, Bukun Huang, Zhizeng Ye, Xuemei Dong, Tuo Chen, Hui Xue, Dingkang Yang, Longtao Huang, Jingqun Tang, and Haiwen Hong. Diffusion probe: Generated image result prediction using CNN probes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 35926–35935, 2026. arXiv:2602.23783.

[5] Dave Epstein, Allan Jabri, Ben Poole, Alexei A. Efros, and Aleksander Holynski. Diffusion self-guidance for controllable image generation. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.00986.

[6] Huan Fu and Guoqing Cheng. Enhancing semantic mapping in text-to-image diffusion via gather-and-bind. *Computers & Graphics*, 125:104118, 2024. doi: 10.1016/j.cag.2024.104118.

[7] Dhruba Ghosh, Hannaneh Hajishirzi, and Ludwig Schmidt. GenEval: An object-focused framework for evaluating text-to-image alignment. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks*, 2023. arXiv:2310.11513.

[8] Paul Grimal, Hervé Le Borgne, and Olivier Ferret. Text-to-image alignment in denoising-based models through step selection, 2025. arXiv:2504.17525.

[9] Xiefan Guo, Jinlin Liu, Miaomiao Cui, Jiankai Li, Hongyu Yang, and Di Huang. InitNO: Boosting text-to-image diffusion models via initial noise optimization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2404.04650.

[10] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross attention control. In *International Conference on Learning Representations*, 2023. arXiv:2208.01626.

[11] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[12] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, 2020. arXiv:2006.11239.

[13] Yushi Hu, Benlin Liu, Jungo Kasai, Yizhong Wang, Mari Ostendorf, Ranjay Krishna, and Noah A. Smith. TIFA: Accurate and interpretable text-to-image faithfulness evaluation with question answering, 2023. arXiv:2303.11897.

[14] Kaiyi Huang, Kaiyue Sun, Enze Xie, Zhenguo Li, and Xihui Liu. T2I-CompBench: A comprehensive benchmark for open-world compositional text-to-image generation. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks*, 2023. arXiv:2307.06350.

[15] Amita Kamath, Kai-Wei Chang, Ranjay Krishna, Luke Zettlemoyer, Yushi Hu, and Marjan Ghazvininejad. GenEval 2: Addressing benchmark drift in text-to-image evaluation, 2025. arXiv:2512.16853.

[16] Tuomas Kynkäänniemi, Miika Aittala, Tero Karras, Samuli Laine, Timo Aila, and Jaakko Lehtinen. Applying guidance in a limited interval improves sample and distribution quality in diffusion models. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2404.07724.

[17] Marvin Li and Sitan Chen. Critical windows: non-asymptotic theory for feature emergence in diffusion models. In *International Conference on Machine Learning*, 2024. arXiv:2403.01633.

[18] Yumeng Li, Margret Keuper, Dan Zhang, and Anna Khoreva. Divide & bind your attention for improved generative semantic nursing. In *British Machine Vision Conference*, 2023. arXiv:2307.10864.

[19] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. SDEdit: Guided image synthesis and editing with stochastic differential equations. In *International Conference on Learning Representations*, 2022. arXiv:2108.01073.

[20] Matthias Minderer, Alexey Gritsenko, Austin Stone, Maxim Neumann, Dirk Weissenborn, Alexey Dosovitskiy, Aravindh Mahendran, Anurag Arnab, Mostafa Dehghani, Zhuoran Shen, Xiao Wang, Xiaohua Zhai, Thomas Kipf, and Neil Houlsby. Simple open-vocabulary object detection with vision transformers. In *European Conference on Computer Vision (ECCV)*, 2022. OWL-ViT; arXiv:2205.06230.

[21] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

[22] Pinelopi Papalampidi, Olivia Wiles, Ira Ktena, Aleksandar Shtedritski, Emanuele Bugliarello, Ivana Kajic, Isabela Albuquerque, and Aida Nematzadeh. Dynamic classifier-free diffusion guidance via online feedback, 2025. arXiv:2509.16131.

[23] Or Patashnik, Daniel Garibi, Idan Azuri, Hadar Averbuch-Elor, and Daniel Cohen-Or. Localizing object-level shape variations with text-to-image diffusion models. In *IEEE/CVF International Conference on Computer Vision*, 2023. arXiv:2303.11306.

[24] Alec Radford, Jong Wook Kim, Chris Hallacy, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021. arXiv:2103.00020.

[25] Royi Rassin, Eran Hirsch, Daniel Glickman, Shauli Ravfogel, Yoav Goldberg, and Gal Chechik. Linguistic binding in diffusion models: Enhancing attribute correspondence through attention map alignment. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.08877.

[26] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. arXiv:2112.10752.

[27] Seyedmorteza Sadat, Jakob Buhmann, Derek Bradley, Otmar Hilliges, and Romann M. Weber. CADS: Unleashing the diversity of diffusion models through condition-annealed sampling. In *International Conference on Learning Representations*, 2024. arXiv:2310.17347.

[28] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021. arXiv:2010.02502.

[29] Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus Stenetorp, Jimmy Lin, and Ferhan Ture. What the DAAM: Interpreting stable diffusion using cross attention. In *Proceedings of the Association for Computational Linguistics*, 2023. arXiv:2210.04885.

[30] Yilun Xu, Mingyang Deng, Xiang Cheng, Yonglong Tian, Ziming Liu, and Tommi Jaakkola. Restart sampling for improving generative processes. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.14878.

[31] Yuxin Zhang, Weiming Dong, Fan Tang, Nisha Huang, Haibin Huang, Chongyang Ma, Tong-Yee Lee, Oliver Deussen, and Changsheng Xu. ProSpect: Prompt spectrum for attribute-aware personalization of diffusion models. *ACM Transactions on Graphics (SIGGRAPH Asia)*, 2023. arXiv:2305.16225.

Table 19: Mean UNet evaluations per image by arm (PARTIAL TEST, EXPLORATORY).

Arm	UNet calls
A1	80.0
A2	80.0
A3	80.0
A4	80.0
A6	80.0

A Additional results

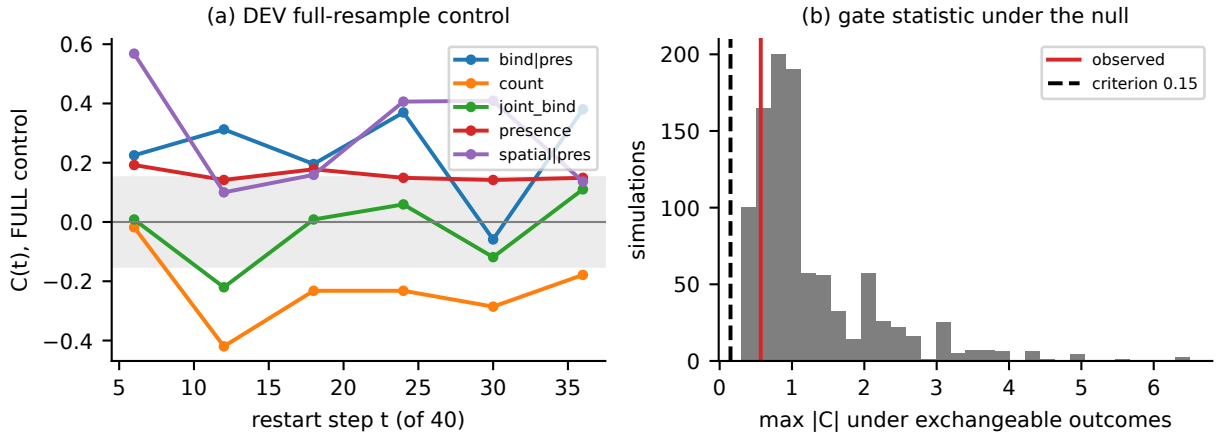


Figure 13: **DEV, post hoc.** (a) Full-resample control: $C(t)$ when the branch is an independent fresh draw (expected 0); grey band: the registered tolerance ±0.15. (b) The registered gate statistic (maximum $|C|$ over types and steps) in 1,000 simulations with exchangeable outcomes and the same design; the observed value (red) is typical of the null, and the criterion (dashed) is essentially never met.

A.1 Additional registered outputs on the partial test split

Supporting variant-level outcome rates (S2), the per-arm network evaluations of the CAG arms, and the equal-ratio restart curves. Every CAG arm evaluates the network 80 times per image, because $w_{lo} = 3$ keeps the unconditional branch.

B Earlier evidence: pilot and infrastructure (exploratory)

This section reports the pilot that motivated the study, (A), the infrastructure and integrity checks of the first cloud runs, (B), and a first-pass look at the DEV core made before the judge audit, (C). None of it tests the registered hypotheses.

B.1 A. Pilot (exploratory)

The pilot generated 1008 images (28 prompts, three CFG scales, two step counts, 6 seeds). Table 21 and Figure 18 show constraint satisfaction; CFG contrasts are in Table 22. With only 4–6 prompts

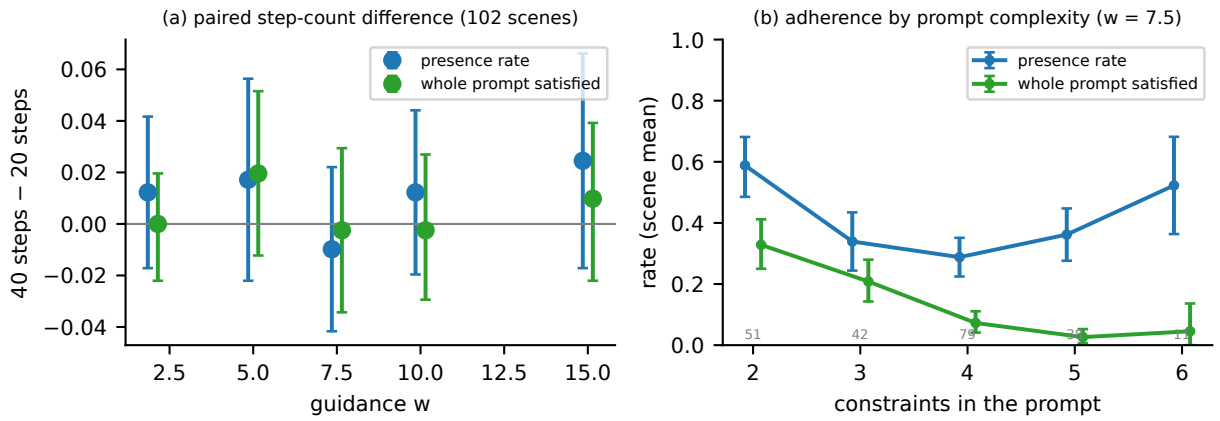


Figure 14: **Partial test split, post hoc.** (a) Scene-paired difference between 40 and 20 DDIM steps in presence rate and whole-prompt satisfaction at each guidance scale (102 scenes; 95% intervals). (b) Adherence at $w = 7.5$ and 40 steps by the number of constraints in the prompt; grey numbers: scenes per group.

Table 20: S2 (EXPLORATORY): outcome rates by prompt variant at $w = 7.5$ on the available TEST shards; 28 rows whose prompt occurs in DEV excluded.

Variant	rows	objects_present	count_ok	bind_given_pres	spatial_given_pres	size_given_pres
CAN	884	0.391	0.257	0.434	0.551	0.494
CF-A	320	0.428	0.362	0.726	–	–
CF-C	314	0.255	0.235	0.237	0.652	0.500
CF-N	148	0.662	0.297	0.622	–	–
CF-O	104	0.981	–	–	–	–
CF-R	166	0.355	–	0.269	0.390	0.222
CF-S	22	0.500	–	0.273	0.818	0.364
ORD	452	0.327	0.132	0.271	0.479	–
PAR	452	0.343	0.277	0.450	0.605	–

per category the intervals are wide and no CFG effect on adherence is resolved; absence of a resolved effect is not evidence of its absence. Global attention re-weighting (scale 2 against the control path) changed overall adherence by +0.02 (95% CI -0.04 to +0.09), i.e. no resolved improvement (Table 23). Among binding-bearing pilot images, all required objects were detected in 395 of 648 (0.61), but only 179 of 432 (0.41) for the binding and multi categories, because counting prompts are always eligible; the pilot's legacy binding score already counts omissions as failures. In prompt-grouped cross-validation, early attention features added ΔAUC of +0.050 (95% CI +0.017 to +0.097) over the design and seed baseline, whereas seed identity added +0.002 (Table 24); only six distinct noise vectors exist in the pilot, a seed-0 deficit appeared (permutation $p = 0.012$), and both are post hoc observations. They motivated an earlier protocol revision's question of whether attention is a forecast or a read-out of the outcome (then labelled HB). The current revision demoted that question to a supporting analysis outside the registered family (HA, HD, HE, HF); Section 11 revisits it descriptively.

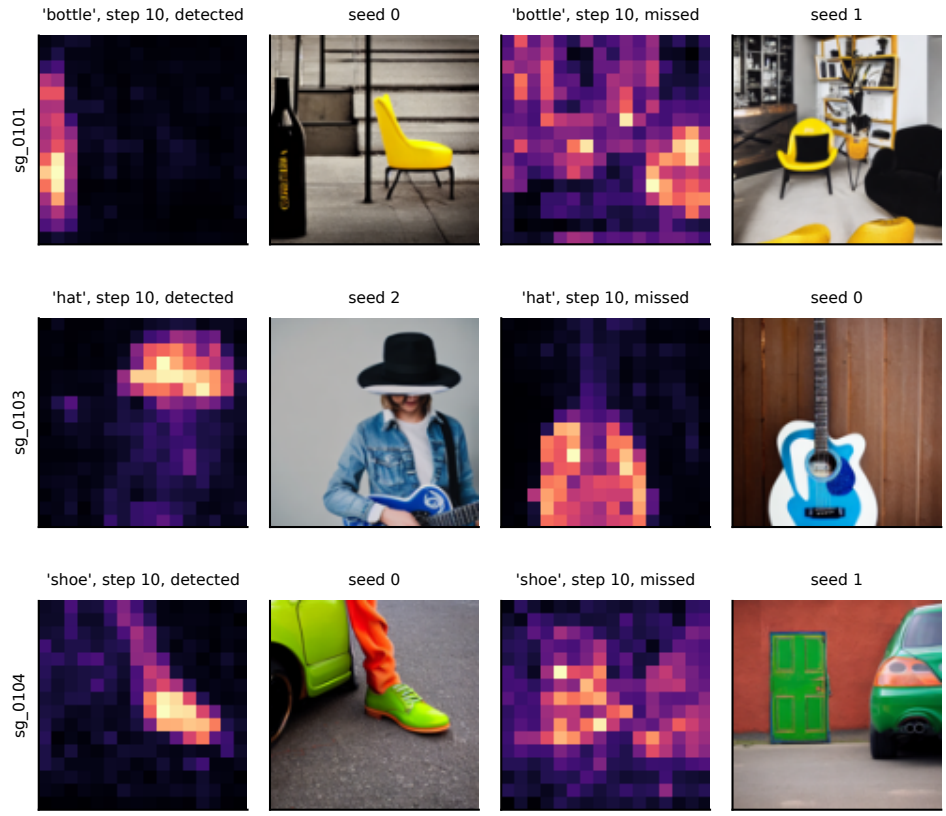


Figure 15: **Illustration, not evidence.** Attention of an object noun at step 10 in a seed where the judge found the object and in a seed where it did not, with the final images ($w = 7.5$). Selection rule (stated in the generating script, independent of appearance): the three lowest main-benchmark scene identifiers with a two-object prompt in which the object is found in some seeds and missed in others; for each, the lowest-index seed of each outcome.

B.2 B. Infrastructure and integrity (not scientific findings)

On a Tesla T4 with the committed code, the GPU identity script ran 3 cases and all six required identities held (determinism, loop from step 0, reload-and-resume, zero-magnitude branch, hook-off control, restored baseline; maximum pixel difference 0). The scope is three pilot prompts at one seed and setting; the model revision is not recorded inside that artifact. The R2a DEV core contains 400 images and 2400 saved checkpoints; an independent audit passed 22 of 22 hard checks (manifest coverage, hashes, readability, no duplicates or orphans, checkpoint layout). A one-sample-pair regeneration on a different GPU matched the stored latent at the same step best, supporting the checkpoint indexing; this was not a bit-level Kaggle comparison. Two throughput probes disagree by about 25% (Table 25) and are not reconciled.

B.3 C. First-pass DEV outcomes (unaudited judge)

Only about 0.40 of binding-bearing DEV images have every required object detected and binding given presence is defined for 113 of 280 images. Whether this reflects the model or the detector

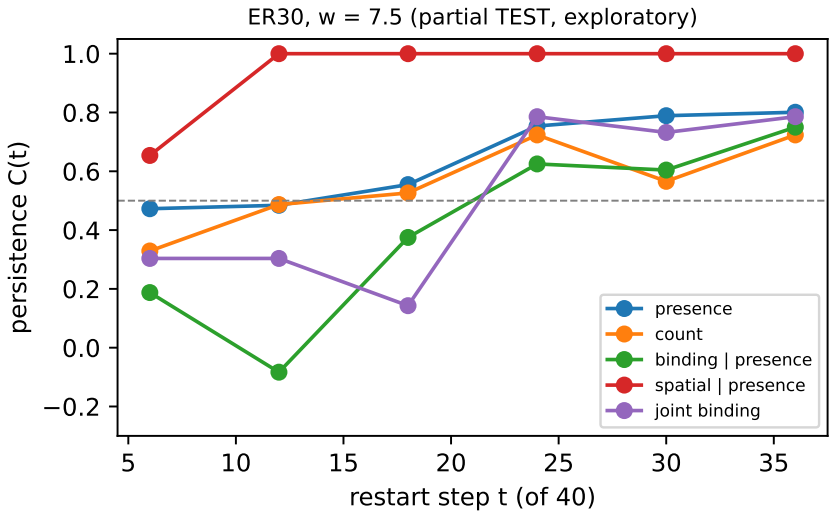


Figure 16: Chance-corrected persistence $C(t)$ by constraint type under the ER30 restart probe ($w = 7.5$); dashed line marks $C = 0.5$. Partial test split, exploratory.

cannot be said before the audit.

C Reproduction details

Every generation, judging and analysis run records its run identifier, source commit, pinned model and judge revisions, resolved configuration, software and GPU environment, and SHA-256 checksums of every output; archives are verified file by file before analysis. The signed protocol binds the hashes of 103 code, configuration, benchmark and documentation files, the judge definition, the development-set constants and the three test-stage manifests, and test generation and the confirmatory analysis refuse to run unless it verifies. The partial-test numbers in this paper come from one verified analysis archive built from source commit `40ade4d`. A second judging of the same images on different hardware reproduced every judged outcome.

Generators, all reading saved evidence only:

- `paper/make_paper_assets.py`: DEV figures, DEV tables and number macros.
- `partial/partial_report.py`: registered analysis on the partial test split.
- `partial/robustness.py`: Section 10.
- `diagnostics/attention.py`: Section 11.
- `diagnostics/full_control.py`: the full-resample diagnosis.
- `scripts/final_report.py`: the registered confirmatory analysis, once all shards exist.

`PAPER_STATUS.md` maps each number in the text to its evidence file and field. Images, latents and human labels are not distributed.

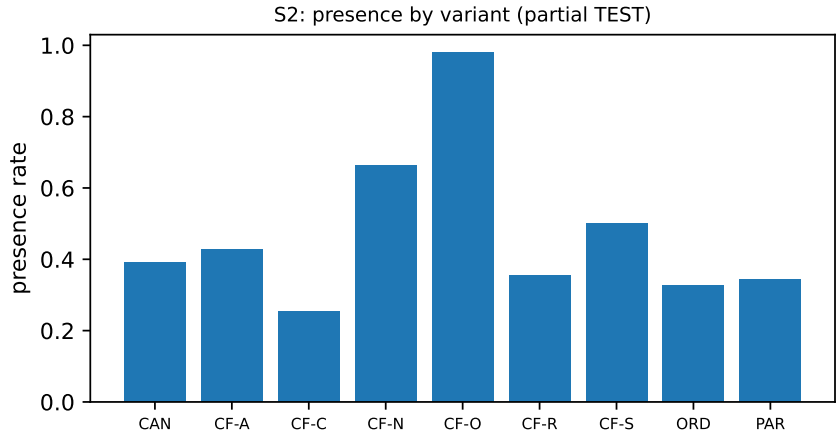


Figure 17: Supporting analysis S2: object-presence rate by prompt variant. Partial test split, exploratory.

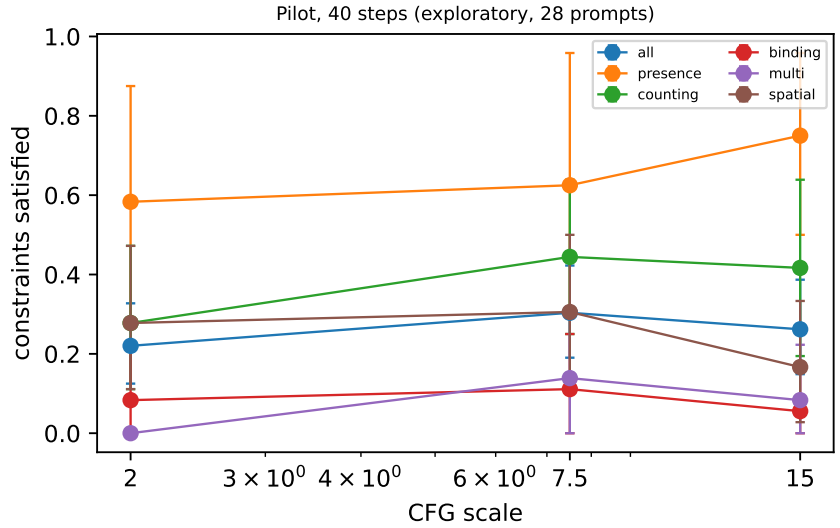


Figure 18: Pilot constraint satisfaction against CFG (40 steps, exploratory).

Table 21: Pilot (28 prompts, detector-based labels, exploratory): fraction of constraints satisfied, by category, CFG and inference steps; 95% prompt-cluster bootstrap intervals.

Category	CFG	20 steps	40 steps	Prompts
all	2	0.24 [0.137, 0.351]	0.22 [0.125, 0.327]	28
all	7.5	0.30 [0.190, 0.417]	0.30 [0.190, 0.423]	28
all	15	0.32 [0.202, 0.440]	0.26 [0.149, 0.387]	28
presence	2	0.58 [0.292, 0.875]	0.58 [0.292, 0.875]	4
presence	7.5	0.71 [0.417, 0.958]	0.62 [0.292, 0.958]	4
presence	15	0.67 [0.292, 1.000]	0.75 [0.500, 0.958]	4
counting	2	0.36 [0.167, 0.583]	0.28 [0.111, 0.472]	6
counting	7.5	0.39 [0.194, 0.611]	0.44 [0.250, 0.639]	6
counting	15	0.44 [0.250, 0.639]	0.42 [0.194, 0.639]	6
binding	2	0.06 [0.000, 0.194]	0.08 [0.000, 0.222]	6
binding	7.5	0.11 [0.000, 0.278]	0.11 [0.000, 0.250]	6
binding	15	0.17 [0.028, 0.333]	0.06 [0.000, 0.167]	6
multi	2	0.00 [0.000, 0.000]	0.00 [0.000, 0.000]	6
multi	7.5	0.11 [0.000, 0.333]	0.14 [0.000, 0.306]	6
multi	15	0.08 [0.000, 0.250]	0.08 [0.000, 0.223]	6
spatial	2	0.31 [0.111, 0.500]	0.28 [0.111, 0.472]	6
spatial	7.5	0.31 [0.139, 0.500]	0.31 [0.111, 0.500]	6
spatial	15	0.33 [0.167, 0.528]	0.17 [0.028, 0.333]	6

Intervals resample prompts; the judge is unaudited on the DEV set.

Table 22: Pilot CFG contrasts (paired by prompt and seed; Holm-adjusted p from Wilcoxon tests), exploratory.

Category	CFG a vs b	Diff.	95% CI	Prompts	p_{Holm}
presence	2 vs 7.5	-0.083	[-0.438, 0.229]	4	1.000
presence	2 vs 15	-0.125	[-0.500, 0.250]	4	1.000
presence	7.5 vs 15	-0.042	[-0.208, 0.104]	4	1.000
binding	2 vs 7.5	-0.042	[-0.181, 0.097]	6	1.000
binding	2 vs 15	-0.042	[-0.153, 0.083]	6	1.000
binding	7.5 vs 15	+0.000	[-0.111, 0.125]	6	1.000
counting	2 vs 7.5	-0.097	[-0.292, 0.083]	6	1.000
counting	2 vs 15	-0.111	[-0.333, 0.083]	6	1.000
counting	7.5 vs 15	-0.014	[-0.097, 0.056]	6	1.000
spatial	2 vs 7.5	-0.014	[-0.167, 0.153]	6	1.000
spatial	2 vs 15	+0.042	[-0.069, 0.125]	6	0.750
spatial	7.5 vs 15	+0.056	[-0.069, 0.181]	6	1.000
multi	2 vs 7.5	-0.125	[-0.292, 0.000]	6	0.750
multi	2 vs 15	-0.083	[-0.222, 0.000]	6	1.000
multi	7.5 vs 15	+0.042	[-0.083, 0.208]	6	1.000
all	2 vs 7.5	-0.071	[-0.155, 0.009]	28	0.174
all	2 vs 15	-0.060	[-0.146, 0.018]	28	0.214
all	7.5 vs 15	+0.012	[-0.045, 0.071]	28	0.873

Table 23: Pilot global attention re-weighting (scale 2 and 4 against the scale-1 control path), primary outcome, exploratory.

Category	Arm	Diff.	95% CI	Prompts	p_{Holm}
presence	rw2 vs control	+0.167	[0.000, 0.375]	4	0.500
presence	rw4 vs control	+0.125	[-0.042, 0.333]	4	0.500
binding	rw2 vs control	+0.028	[-0.111, 0.167]	6	1.000
binding	rw4 vs control	+0.028	[-0.139, 0.222]	6	1.000
counting	rw2 vs control	+0.028	[0.000, 0.111]	6	1.000
counting	rw4 vs control	+0.028	[-0.083, 0.139]	6	1.000
spatial	rw2 vs control	-0.028	[-0.194, 0.139]	6	1.000
spatial	rw4 vs control	+0.056	[-0.139, 0.250]	6	1.000
multi	rw2 vs control	-0.028	[-0.194, 0.111]	6	1.000
multi	rw4 vs control	-0.083	[-0.222, 0.000]	6	1.000
all	rw2 vs control	+0.024	[-0.042, 0.089]	28	0.757
all	rw4 vs control	+0.024	[-0.054, 0.095]	28	0.757

Control images were pixel-identical to the unmodified pipeline (168 of 168 images).

Table 24: Pilot nested prediction of the primary outcome (40-step images, prompt-grouped cross-validation, 28 prompts); exploratory.

Contrast	ΔAUC	95% prompt-bootstrap CI
attn_minus_design	+0.067	[0.028, 0.122]
noise_minus_design	+0.002	[-0.028, 0.027]
attn_minus_noise	+0.050	[0.017, 0.097]

Base rate 0.236. Only six distinct noise vectors exist in the pilot, so noise features equal seed identity (noise vs design: -0.007).

Table 25: Measured throughput on Tesla T4 (seconds per 40-step image, first image excluded) from two probe sessions; the real R2a run gave a median of 6.21 and 6.26 s per worker.

Session	Plain	With latents	Two workers, wall (s)
R1 session 10:10 UTC	6.16	6.25	53.9
R0 session 15:02 UTC	7.76	7.59	60.5

Table 26: R2a DEV core, first-pass detector (J1) outcomes on binding-bearing canonical prompts. **Unaudited judge; exploratory.**

Quantity	Value	n
All required objects detected	0.404	280
Binding given presence	0.487	113
Joint (present and bound)	0.196	280

70 scenes × 4 seeds. These outcomes precede the judge audit (§6).